

GENERAL SCIENCE GRADE 10: TRANSPORT IN PLANTS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.4 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 1.0: Life Science
Sub-Strand	Sub-Strand 1.4: Transport in Plants
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Absorption of Water and Mineral Salts – Translocation – Factors affecting transpiration – Importance of transpiration
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe absorption of water and mineral salts in plants, - explain translocation in plants, - investigate factors affecting transpiration in plants, - describe the significance of transpiration in sustainability of plant life, - appreciate the role of transport in plants.
Core Competencies	– Citizenship: as the learner sensitises the community on the importance of watering and manuring plants to sustain plant growth. – Learning to learn: as the learner reflects on experiments on factors affecting transpiration in plants.
Core Values	– Unity: as the learner engages in assigned roles while working with others in the process of investigation of factors affecting transpiration. – Patriotism: as the learner sensitises the community on the importance of watering and manuring plants to sustain plant growth.
Science & Engineering Practices	– Planning and Carrying Out Investigations: as the learner designs and conducts experiments to investigate factors affecting transpiration in plants (e.g., leaf area, wind, humidity, light intensity). – Analysing and Interpreting Data: as the learner records, compares, and draws conclusions from transpiration experiment results across different environmental conditions. – Constructing Explanations: as the learner uses evidence from absorption and translocation experiments to explain how water and mineral salts move through a plant. – Obtaining, Evaluating, and Communicating Information: as the learner uses digital devices and print media to research adhesive and cohesive forces involved in transport, then shares findings with peers.

Pertinent & Contemporary Issues (PCIs)	– Safety and security: as the learner observes safety precautions while carrying out experiments in groups to investigate factors affecting transpiration in plants. – Environmental issues: as the learner creates awareness in the community on the use of fertilisers (Di-ammonium Phosphate, Calcium Ammonium Nitrate, Urea) in boosting crop production.
Career Connections	– Agriculture / Agronomy: understanding water and mineral salt absorption directly informs irrigation scheduling, fertiliser application, and crop management on Kenyan farms in areas like the Rift Valley and Kericho. – Botany / Plant Science: knowledge of translocation and transpiration underpins plant physiology research and breeding programmes at institutions such as KALRO (Kenya Agricultural and Livestock Research Organisation). – Environmental Science: understanding transpiration and water loss links to watershed management, agroforestry, and conservation of water catchment areas around Mt. Kenya and the Mau Forest. – Horticulture / Floriculture: flower and vegetable export farms around Naivasha require precise knowledge of plant water transport to maintain produce quality for international markets. – Biomedical / Pharmaceutical Science: principles of transport across membranes underpin drug delivery and IV fluid therapy studied in nursing and medical training.
Focus for Lessons	This unit centres on how plants absorb, move, and lose water and dissolved substances — processes critical to crop survival across Kenya's diverse agro-ecological zones. Learners investigate absorption of water and mineral salts through roots, translocation of organic and inorganic solutes through the phloem and xylem, and the environmental factors that drive or limit transpiration. The unit builds from observable phenomena on a school farm or garden toward molecular-level explanations of cohesion-tension and source-to-sink transport, equipping learners to make evidence-based decisions about watering and fertilising crops in their local communities.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end? KICD KEY INQUIRY QUESTION: 1. Why is transport in plants important?
Anchoring Phenomenon	A freshly uprooted maize seedling from the school garden in Kisumu is placed with its roots in a beaker of water coloured with red food dye. After 30 minutes, when the stem is cut cross-sectionally, vivid red rings appear in the woody tissue — but the rest of the stem remains white. At the same time, a leafy branch of a succulent plant (e.g., a bean or sweet potato cutting) sealed in a clear polythene bag left in the afternoon sun shows a film of water droplets forming on the inside of the bag within minutes, even though no liquid water was added to the bag. Both observations are puzzling: How does coloured water travel upward against gravity through only certain channels in the stem? And where does all that water vapour inside the bag actually come from — and why does the plant seem to be "losing" water even when it needs it to survive?
Storyline Thread	Lesson 1 — PREDICT Anchoring Phenomenon Introduction: Students observe the dyed-water maize seedling and the sweating polythene-bag demonstrations. Working in pairs, they sketch where they think the red dye will appear in a cross-section and predict where the water droplets in the bag come from. Predictions are posted on the Driving Question Board (DQB). Teacher introduces the driving question and records all student wonder questions. [SEP: Asking Questions] Lesson 2 — OBSERVE Structures for Absorption: Learners observe prepared slides or photomicrographs of root hair cells and cross-sections of roots and stems. They identify xylem and phloem using labelled diagrams and digital images. Teacher demonstrates the adhesive and cohesive forces using a capillary tube in water. Students record observations and revisit their Lesson 1 predictions. [SEP: Obtaining and Evaluating Information; Constructing Explanations]

	Lesson 3 — EXPLAIN Absorption of Water and Mineral Salts: Guided discussion on osmosis, active transport, and the root pressure mechanism. Students use print/digital media to research how mineral salts (e.g., DAP — Di-ammonium Phosphate) enter root hair cells and are carried upward. Kenyan context: link to fertiliser use on maize and sukuma wiki farms. Students update the DQB with new explanations. [SEP: Constructing Explanations] Lesson 4 — OBSERVE & EXPLAIN Translocation: Learners carry out or observe a demonstration experiment on translocation (e.g., ring-barking/girdling a soft stem, or observing phloem exudate from a cut surface). They discuss the source-to-sink model of phloem transport and compare xylem vs. phloem function using a two-column chart. Peers share findings in plenary. DQB updated with translocation evidence. [SEP: Analysing and Interpreting Data; Constructing Explanations] Lesson 5 — OBSERVE Transpiration Experiment Setup: Groups design and set up experiments to investigate factors affecting transpiration — varying one factor each: (a) leaf surface area covered with petroleum jelly, (b) fan creating wind, (c) polythene bag enclosure raising humidity, (d) varying light intensity. Each group states a hypothesis, identifies variables, and records initial measurements. Safety precautions reviewed. [SEP: Planning and Carrying Out Investigations] Lesson 6 — OBSERVE & ANALYSE Transpiration Data Collection and Analysis: Groups collect final transpiration measurements, record data in tables, and draw bar/line graphs. Class data is pooled on the board; groups compare results and identify patterns. Teacher facilitates sensemaking: why does wind increase transpiration? How does humidity affect the rate? Link back to the polythene-bag phenomenon from Lesson 1. [SEP: Analysing and Interpreting Data; Obtaining, Evaluating, and Communicating Information] Lesson 7 — EXPLAIN Importance of Transpiration and Community Application: Whole-class discussion on the significance of transpiration: cooling of leaves, creation of the transpiration pull (cohesion-tension theory), and mineral salt distribution. Students research and discuss the environmental importance of transpiration for local water cycling (e.g., Mau Forest catchment). Community action component: students draft a short awareness message for farmers on proper watering, manuring, and responsible fertiliser use (DAP, CAN, Urea). DQB revisited and updated. [SEP: Constructing Explanations; Obtaining, Evaluating, and Communicating Information] Lesson 8 — MODEL BUILDING & DQB CLOSURE Cumulative Model Revision and Assessment: Students individually revise their Lesson 1 sketches into a complete annotated diagram showing the full journey of a water molecule from soil → root hair → xylem → leaf → stomata → atmosphere, annotating each step with the mechanism (osmosis, cohesion-tension, transpiration pull, active transport for minerals). Groups present their models; peers provide feedback. DQB is reviewed: which questions are now answered? Which remain open? Teacher administers a short performance task (explain why a wilting tomato plant in Mombasa's coastal heat recovers after watering at dusk). [SEP: Developing and Using Models; Constructing Explanations]
Supporting Phenomena	– A freshly cut celery stalk placed in coloured water overnight reveals dyed streaks running up through the outer ring of the stalk, making the vascular bundles visible to the naked eye — raising the question of what structures are responsible and how water climbs so high. – A potted bean plant (common in Kenyan school gardens) left unwatered for two days wilts dramatically; when watered, it recovers within hours — demonstrating that water transport is directly linked to plant structural support (turgor), not just nutrition. – A banana farmer in the Rift Valley applies Di-ammonium Phosphate (DAP) fertiliser to the soil, and within days the leaves become noticeably greener and more turgid — prompting questions about how dissolved mineral salts reach leaves from soil,

and what happens if too much fertiliser is applied.

- On a hot, dry afternoon in Nairobi, the soil under a shady tree feels noticeably cooler and slightly moist compared to bare soil nearby — suggesting that the tree is releasing water vapour through its leaves, with implications for local microclimate and water cycling.

LESSON 1 (40 minutes): Anchoring Phenomenon: The Travelling Red Dye & The Sweating Bag

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the sub-strand by anchoring learners in two vivid, puzzling observations — dyed water moving upward through specific channels in a maize stem, and water vapour forming inside a sealed polythene bag around a leafy cutting — so that learners generate genuine questions that will drive the entire Transport in Plants inquiry.
Knowledge	Learners will know that (1) water and mineral salts enter plants through roots and move upward through specific vascular tissue, and (2) plants release water vapour through their leaves even when no liquid water is visibly present on them.
Skills	Learners will be able to: (a) make and record evidence-based predictions using labelled diagrams; (b) ask testable questions about an observed phenomenon (SEP: Asking Questions); (c) record initial ideas on a Driving Question Board (DQB).
Attitudes	Learners appreciate that careful observation of everyday Kenyan farm plants can reveal deep scientific puzzles worth investigating, and develop curiosity and open-mindedness towards unexpected results.
Key Inquiry Question	Why is transport in plants important?
Purpose in Storyline	Lesson 1 opens the storyline. Learners encounter both dimensions of the driving question simultaneously — upward movement of water through hidden channels (translocation/xylem) and the mysterious loss of water from leaves (transpiration) — without yet receiving any explanation. Their unanswered questions and prediction diagrams seed the DQB and create the intellectual need that will be resolved across subsequent lessons.
Safety Notes	Red food dye is non-toxic but can stain clothing — learners should wear aprons or old clothes. Warn learners not to taste any coloured water. When cutting stems cross-sectionally, the teacher (or a designated learner under close supervision) should use the razor blade on a cutting board away from fingers. Polythene bags should not be placed over faces. Dispose of dyed water responsibly — do not pour into the school garden.

B. LESSON OVERVIEW

Learners gather around two teacher-led demonstrations: (1) a freshly uprooted maize seedling from the school garden, its roots submerged in water coloured with red food dye, and (2) a leafy sweet potato or bean cutting sealed inside a clear polythene bag placed in a sunny spot. Before any explanation is given, learners individually sketch cross-sections of the maize stem to predict where the red dye will appear, and they predict the origin of the water droplets forming on the inside of the polythene bag. Pairs share predictions, then predictions are posted on the newly created Driving Question Board. The teacher introduces the formal driving question and harvests all learner wonder-questions onto the DQB. No answers are given in this lesson — the goal is to open productive uncertainty. The lesson uses SEP: Asking Questions throughout.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown the two set-up demonstrations BEFORE any	1. Hold up the maize seedling (roots in the beaker of red water)	THINK-PAIR-SHARE WITH PREDICTION DIAGRAMS —	Observable indicator: Every learner has a shaded cross-

 VIDEO: WATCH: The Agricultural Revolution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	results are visible (the dyed-water maize seedling has been soaking for only 5 minutes at this point; the polythene bag has just been sealed around the cutting and placed in sunlight). Working individually for 3 minutes, each learner draws a blank circle (representing a cross-section of the maize stem) in their exercise books and shades in where they predict the red dye will appear after 30 minutes. They also write one sentence answering: 'Where do you think the water droplets inside the polythene bag come from?' Learners then compare their prediction diagrams with a partner (Think-Pair) for 2 minutes, noting agreements and disagreements.	and the sealed polythene bag side by side. Say exactly: 'I uprooted this maize plant from our school garden here in Kisumu this morning. I have put its roots in coloured water. I have sealed this sweet potato cutting in a clear bag and placed it in the sun. Do NOT tell me what will happen — I want YOUR ideas first.' 2. Distribute blank cross-section circle templates (pre-drawn on paper) OR instruct learners to draw a 5 cm circle in their books. Say: 'In this circle, shade the area where you think the red colour will appear when I cut the stem across. There is no wrong answer right now.' 3. Allow 3 minutes of silent individual thinking. WAIT TIME: do not speak or prompt during these 3 minutes. 4. Say: 'Now turn to your partner. Compare your diagrams. Agree on one shared prediction and be ready to explain your reasoning.' Allow 2 minutes. 5. Cold-call exactly 4 pairs (select pairs with visibly different predictions). Ask each pair: 'Show us your diagram and tell us — WHY did you shade that part?' After each response, say: 'Interesting. Hold that thought.' Do NOT confirm or correct. 6. For the polythene bag: cold-call 3 individual learners. Ask: 'Amina — where do you think that water vapour is coming from?' WAIT 12 seconds after asking before accepting any answer.	Learners externalise their mental models by drawing before they have data. This reveals prior conceptions (e.g., 'the whole stem will turn red' vs. 'only the outer ring') and creates a personal stake in finding out whether they were right. The diagram format makes abstract internal anatomy concrete and comparable.	section diagram and a written sentence in their book before pairs share. Teacher scans books during the pair-share to identify: (a) learners who shade the entire circle (no prior knowledge of vascular tissue), (b) learners who shade a ring or spots (partial prior knowledge), (c) learners who draw dots or a central column (alternative conceptions). Teacher mentally notes 2–3 learners from group (a) to revisit in Lesson 3 (Explain phase on xylem).
Observe Phase  VIDEO: WATCH: The	After approximately 25 minutes of lesson time have elapsed (the maize seedling has been soaking for ~30 minutes total including	1. Build anticipation: 'It has now been about 30 minutes. Let us see what happened.' Carry the maize seedling to the cutting board at the	ANNOTATED OBSERVATION DIAGRAMS WITH PREDICTION COMPARISON — By drawing what they predicted AND then	Observable indicator: Learner books show two layers of annotation — original prediction (one colour) and observed result

Agricultural Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	setup time before class), the teacher performs the cross-sectional cut live in front of the class. Learners observe the vivid red rings appearing only in the woody vascular tissue. They record what they see in a labelled diagram, writing at least two observations (e.g., 'The red colour appears only in small dots/rings, not in the whole stem' and 'The outer green part and the central pith stayed white'). Simultaneously, learners observe water droplets that have formed on the inside of the polythene bag around the sweet potato or bean cutting. Learners record two observations about the bag ('Droplets formed on the inside surface of the bag' and 'The leaves look the same — no visible water on the outside of the leaves'). Learners annotate their original prediction diagrams with a different-coloured pen to show what they actually observed.	front. Say: 'Watch carefully — I will cut across the stem here' (indicate a point about 5 cm above the roots). Make a clean cross-sectional cut with a sharp razor blade on the cutting board. Hold the cut surface up so all learners can see the red rings. 2. Say: 'Write down exactly what you see. Not what you think — what you SEE.' Allow 90 seconds of silent recording. 3. Walk the cut stem around to at least 3 learner pairs for close-up observation. Say: 'Look at the pattern. Can you see any difference between the red parts and the white parts?' WAIT 10 seconds. 4. Hold up the polythene bag. Say: 'Now look at this bag. I sealed a sweet potato cutting inside it 30 minutes ago. No liquid water was added. What do you observe?' Cold-call 3 learners. After each: 'Write that observation down.' 5. Instruct: 'Go back to your prediction diagram. Using a RED pen or pencil, draw what you actually observed. Then put a tick if your prediction was right, or a cross if it was different — and write one word describing how you feel about the difference.' 6. Ask the class: 'How many of you predicted exactly where the red appeared?' (Show of hands — count aloud.) 'How many were surprised?' (Count aloud.) Record counts on the board: e.g., 'Predicted correctly: 4 Surprised: 28'. Leave these numbers on the board for the DQB phase.	annotating with what they observed using a contrasting colour, learners perform a cognitive comparison that highlights the gap between their prior model and the evidence. The surprise data (only 4 correct out of 32) becomes a shared class experience that motivates explanation-seeking.	(contrasting colour/pen). Teacher checks that observations are descriptive, not explanatory (e.g., 'red rings in the woody dots' is acceptable; 'because water went up the xylem' is premature — teacher gently redirects: 'That is an explanation — for now, just tell me what you see'). At least 80% of learners should have two distinct observation statements recorded before the class moves on.
Explain Phase  VIDEO: WATCH: The	Learners attempt to construct initial explanations using only the evidence they have just observed	1. Write the two questions on the board in large print before the group work begins. Read them	COLLABORATIVE EVIDENCE-BASED ARGUMENTATION (Initial) — Groups are required to	Observable indicator: Each group has written at least 2–3 sentences that reference specific

Agricultural Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	— no textbook input yet. In groups of four, learners answer two questions written on the board: (Q1) 'Why do you think the red dye appeared ONLY in those specific parts of the stem — and not everywhere?' (Q2) 'If no liquid water was added to the bag, where could the water vapour have come from?' Groups have 5 minutes to discuss and write a group explanation (2–3 sentences per question) on a strip of manila paper or in their books. Groups share one idea each in a whole-class share-out. Teacher scribes key words from each group on the board under two columns: 'Maize Stem — Dye Ideas' and 'Bag — Water Vapour Ideas'. Teacher does NOT evaluate or correct any idea at this stage.	aloud. Say: 'You do not need to know the correct science answer yet. I want YOUR best explanation based only on what you just observed. Use your diagrams as evidence.' 2. Circulate during the 5 minutes of group work. Listen to group discussions. Prompt quieter groups: 'What did you notice about WHERE the red rings were? Does that give you a clue?' WAIT 12 seconds after each prompt before speaking again. 3. After 5 minutes, call the class to attention. Take exactly one idea from each of the 6–8 groups (no repeats — if a group has the same idea as a previous group, say 'You agree with Group 3 — noted. Does your group have a different idea?'). 4. Scribe their exact words — do not paraphrase or correct. Underline key terms learners use (e.g., 'tubes', 'pipes', 'channels', 'skin of the leaf', 'breathing'). 5. When all groups have shared, step back and say: 'Look at this board. We have several different ideas. They cannot all be right — but they might all point at something true. The experiments in the next few lessons will help us figure out which ideas hold up.' 6. Say: 'I want to add one word to our vocabulary today — not to explain yet, but so we have a name for the question.' Write 'TRANSPORT IN PLANTS' on the board. 'This sub-strand is about exactly these two mysteries.'	ground their explanations in the specific observations they recorded (the diagram evidence), not in recalled knowledge. By scribing all ideas without evaluation, the teacher creates a 'public record of current thinking' that the class will return to and revise across the sub-strand — making learning visible as a process of refinement rather than a single transfer of correct information.	observations (e.g., 'We think there are special tubes because the red only went to the dots, not the white parts'). Teacher listens for learners who invoke osmosis, capillary action, or evaporation by name — these learners have relevant prior knowledge and can be paired as peer resources in later lessons. Teacher notes on a class list any groups producing only one-word answers ('pipes') with no reasoning — these groups will receive a sentence-starter scaffold in Lesson 2.
Driving Question Board (DQB) Creation  VIDEO:	The class creates its first Driving Question Board together. The teacher reads the formal driving question aloud: 'How does a maize plant on a Kenyan farm move	1. Unroll the large manila paper DQB and fix it to the wall. Write the driving question at the top in large bold letters. Say: 'This is our Driving Question Board.'	DRIVING QUESTION BOARD — The DQB externalises the class's collective curiosity and creates a living document of the inquiry. By physically mapping questions to	Observable indicator: Every learner has posted at least one question. Teacher reviews question quality after class — questions should be investigable

WATCH: The Agricultural Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?' Each learner writes ONE question of their own — something they genuinely want to know after today's observations — on a sticky note or small piece of paper. Questions are posted on the DQB (a large sheet of manila paper or a section of the classroom wall). The teacher groups the posted questions into clusters in real time: questions about HOW water moves up, questions about WHERE it goes, questions about WHY the plant does this, and questions about WHAT the coloured parts were. The DQB stays on the wall for the entire sub-strand.	Everything we learn in this sub-strand will help us answer this one big question.' 2. Hand each learner a sticky note or a small cut piece of paper. Say: 'Write ONE question — your most burning question from what you saw today. It must be a question you genuinely do not know the answer to. Write your name on the back.' Allow 2 minutes of silent writing. 3. Learners come up row by row to post their questions. As they post, teacher reads each question aloud and asks: 'Should this go near the HOW water moves cluster, or the WHY cluster, or somewhere else?' Let the class decide placement. 4. After all questions are posted, step back and say: 'Count with me — how many questions have we just created?' Count aloud together. 'All of these are worth investigating. By the end of this sub-strand, we will be able to answer every single one of them.' 5. Point to specific questions and say: 'This question — Wanjiku's question about why the dye only went to some parts — we will specifically come back to in Lesson 2. This one about the bag — Otieno's question — we tackle in Lesson 4 on transpiration.' 6. Take a photograph of the DQB (if a phone or tablet is available) for the ARES platform record. Say: 'We will add to this board every lesson. Do not give me the answers yet — let the experiments do that.'	future lessons, the teacher makes the sub-strand structure transparent and gives learners agency over the direction of learning. Returning to the DQB each lesson to tick off answered questions reinforces the cumulative, evidence-building nature of scientific inquiry (SEP: Asking Questions).	(e.g., 'What are those dot-shaped parts in the stem?') rather than purely factual-recall ('What is xylem?'). Any learner who writes only a factual-recall question receives a verbal prompt in the next lesson: 'Your question asks for a name — can you turn it into a HOW or WHY question?' Teacher records the total number of questions and their cluster distribution as a baseline measure of prior curiosity and conceptual gaps.
Model Building Phase VIDEO:	As the final activity, each learner draws an 'Initial Model' in their exercise book — a simple diagram of a maize plant showing: (a) the	1. Say: 'Before we finish today, I want you to draw your first model. A model in science is not a perfect picture — it is your current best	CUMULATIVE INITIAL MODEL — By building a personal model on Day 1 before any instruction, learners commit their prior	Observable indicator: Every learner's book has a dated diagram with at least one arrow showing water movement and at

WATCH: The Agricultural Revolution Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	roots in the soil near Lake Victoria's shore or a Kisumu farm, (b) the stem, and (c) the leaves at the top. On this diagram, learners draw arrows to show their CURRENT best idea of how water moves from the soil to the topmost leaf, and where they think water 'ends up'. Learners also label anything they are uncertain about with a '?' symbol. These models are NOT collected or corrected — they are dated and kept in the learner's own book to be revised in every subsequent lesson as understanding grows.	thinking, shown as a diagram.' 2. Draw a simple outline of a maize plant on the board (roots, stem, 3–4 leaves). Say: 'Draw something like this in your book. Now add arrows showing where you think water goes. Use a '?' anywhere you are not sure.' 3. Allow 4 minutes of silent individual drawing. Do not assist — say: 'I want YOUR thinking, not mine.' 4. Ask 2 learners (cold-call — choose one high-confidence and one quieter learner) to briefly describe their model to the class without showing it: 'Kamau, tell us — where did you draw your arrows going, and why?' WAIT 10 seconds. After Kamau: 'Fatuma, is your model similar or different? What is one thing you marked with a question mark?' 5. Say: 'Put today's date next to your model and write the words INITIAL MODEL underneath. We will come back to this diagram in every lesson and add to it or change it. By Lesson 5, this same page will look very different — and that change IS the learning.' 6. Close the lesson: 'Today you observed something that has puzzled scientists for centuries. You made predictions, you observed real evidence, you shared ideas, and you asked great questions. You did exactly what scientists do. Next lesson, we go into the laboratory to find out what those ring-shaped structures in the maize stem actually are.'	conceptions to paper in a revisable form. The explicit instruction to mark uncertainties with '?' normalises not-knowing as a productive scientific state. Dating the model creates a longitudinal record of conceptual change that learners can celebrate at the sub-strand's end — making the growth of understanding visible and personally meaningful.	least one '?' label. Teacher circulates and notes: (a) learners who draw water moving in a continuous loop (evaporation from leaves → rain → soil → roots → leaves) — these learners have an intuitive sense of the water cycle and will find transpiration meaningful in Lesson 4; (b) learners who draw water moving only to the roots (no upward movement) — these learners need the Lesson 2 xylem evidence most urgently; (c) learners with no arrows at all — teacher writes a brief encouraging note in their book: 'Your question mark shows great scientific honesty. Lesson 2 will give you your first clue.'
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D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) How many learners posted investigable HOW/WHY questions on the DQB versus simple factual-recall questions? If more than 40% posted recall questions, open Lesson 2 with a 5-minute DQB question-refinement activity. (2) Were learners genuinely surprised by the location of the red dye, or did several predict vascular rings correctly? If prior knowledge is higher than expected, increase the complexity of the Lesson 2 observation task. (3) Did any learner connect the polythene bag observation to the concept of sweating or breathing in animals? Note these learners — their analogy can be used in the Lesson 4 transpiration discussion. (4) Was the razor-blade cutting demonstration safe and clearly visible to all learners? If learners at the back could not see the cross-section, plan to prepare 3–4 pre-cut stem sections for Lesson 2 so every pair can examine one up close. (5) Review the DQB question clusters — if no learner asked a question about the white (phloem) tissue or about nutrients (as distinct from water), introduce a teacher-planted question card on the DQB: 'What about the white parts of the stem — do they do anything?' to seed the translocation discussion in Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A maize seedling from the Kisumu school garden with its roots in red-dyed water: after 30 minutes, a cross-sectional cut revealed vivid red rings only in specific dot-shaped woody regions of the stem — the outer green tissue and central pith remained completely white. Simultaneously, water droplets formed on the inside of a sealed polythene bag around a sweet potato cutting placed in afternoon sun, even though no liquid water had been added to the bag.
What did I learn?	Water does not move randomly through the entire stem — it travels through specific, distinct channels. The leaves of a plant release water vapour even when the plant appears dry on the outside. These two observations together suggest that plants actively move water in a directed pathway from roots to leaves, and that water exits the plant through the leaves in a form we cannot see until it condenses.
How does this explain the phenomenon?	No full explanation yet — this lesson opens the inquiry. Learners have generated initial models and posted their questions on the Driving Question Board. The key questions to be resolved across the sub-strand are: What are those ring-shaped channels and how do they pull water upward against gravity? Why does the plant release water through its leaves even when it needs water to survive? These questions will be answered through the laboratory and observation activities in Lessons 2–5.

LESSON 2 (80 minutes): Structures for Absorption: Root Hair Cells, Xylem & Phloem, and Capillary Forces

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather direct observational evidence — from photomicrographs of root hair cells and stem cross-sections, and from a live capillary-tube demonstration — to identify the specific structures and physical forces responsible for water movement in a maize plant, directly revisiting their Lesson 1 predictions.
Knowledge	Learners describe the structure and function of root hair cells; identify xylem and phloem in root and stem cross-section diagrams and photomicrographs; and explain how adhesive and cohesive forces drive water movement through narrow tubes against gravity.
Skills	Obtaining and Evaluating Information (SEP): learners examine labelled photomicrographs and diagrams to identify xylem and phloem. Constructing Explanations (SEP): learners connect capillary-rise evidence to the red-dye observation from the phenomenon. Learners also practise scientific drawing of cell structures with accurate labels.
Attitudes	Learners appreciate that careful structural observation — as practised by Kenyan agricultural scientists studying crop water-use efficiency — is the foundation for explaining how transport works, and develop curiosity about how invisible physical forces do biological work.
Key Inquiry Question	Why is transport in plants important?
Purpose in Storyline	In Lesson 1, learners recorded predictions about how coloured water travels upward through a maize stem and where the water vapour in the sealed polythene bag comes from. In this OBSERVE lesson, learners gather the structural and physical evidence needed to evaluate those predictions: they see which tissue the red dye actually stains (xylem), understand why xylem tubes are arranged in rings, and watch water climb unaided in a capillary tube — setting the stage for Lesson 3's mechanistic explanation of the cohesion-tension theory.
Safety Notes	Food dye solutions are non-toxic but can stain clothing — learners should wear aprons or old clothes. Glass capillary tubes are fragile; handle with care and place in the foam holder provided. Do not touch cut stem surfaces with bare hands if thorns or sharp pith edges are present. Dispose of used dye solution as directed by the teacher. No open flames in this lesson.

B. LESSON OVERVIEW

Learners begin by revisiting their Lesson 1 prediction cards about the maize-stem red-dye and polythene-bag phenomena. The teacher then guides three sequential evidence-gathering activities: (1) structured observation of photomicrographs of root hair cells and root/stem cross-sections to identify and label xylem and phloem; (2) a live teacher demonstration of capillary rise using glass tubes of different diameters in water coloured with food dye, drawing the explicit parallel to the narrow xylem vessels seen in the cross-sections; and (3) a gallery walk in which pairs examine six large labelled diagrams posted around the classroom and complete an observation recording sheet. The lesson closes with a whole-class evidence-consolidation discussion in which learners annotate their Lesson 1 prediction cards with new observational evidence, preparing the class Driving Question Board for Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners retrieve their Lesson 1 prediction cards (or exercise-book	Display the phenomenon photo on the projector or a large printed	Prediction Revisitation — by explicitly returning to prior	Teacher listens to the 3 cold-called pairs' responses and mentally

VIDEO: Intro to exponential functions Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Geometric Visual of a Square Root (PLIX) Source: CK-12 Search ARES for similar readings	predictions) about: (a) which part of the maize stem the red dye travelled through, and (b) where the water vapour in the sealed polythene bag came from. Working in pairs for 3 minutes, they re-read their prediction and write one sentence beginning: 'I still think... / I am now less sure about... because...'. Three pairs are cold-called to share aloud. Learners then silently examine the phenomenon photograph on the board — the cross-sectional cut of the red-stained maize stem — for 60 seconds before the teacher poses the focus question.	poster (A2 minimum). Say exactly: 'Look at this cut maize stem from our Kisumu school garden. The red rings appeared only in one type of tissue. Before we find out which tissue, re-read your Lesson 1 prediction card. You have 3 minutes with your partner — go.' [Allow 3 minutes.] Cold-call 3 pairs using the name-stick method. After each response, say: 'Thank you — hold that thought. We are going to gather evidence today that will help you decide whether to keep or revise that prediction.' Post the focus question on the board: 'What structures inside a maize stem carry water upward — and what physical forces make that possible?' Allow 10 seconds of silent reading (WAIT TIME).	predictions before new input, learners activate relevant prior knowledge and create a cognitive 'gap' (Bransford, 2000) that motivates observation. Connecting to the phenomenon anchors the new vocabulary (xylem, phloem, cohesion, adhesion) to a concrete, visible image rather than an abstract definition.	notes: Do learners name a specific tissue (e.g., 'the outer ring', 'the coloured part'), or do they use vague language ('it just goes up')? This indicates depth of prior structural understanding. Teacher uses this to calibrate how much scaffolding to provide during Activity 1.
Observe Phase VIDEO: Intro to exponential functions Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Geometric Visual of a Square Root (PLIX) Source: CK-12 Search ARES for similar readings	ACTIVITY A — Photomicrograph Analysis (20 min): Each pair receives Handout 2A, which contains: (i) a labelled photomicrograph of a root hair cell (showing the elongated hair, thin cell wall, large vacuole, and absence of chloroplasts); (ii) a transverse section (TS) of a root showing the epidermis, cortex, endodermis, and vascular bundle with xylem and phloem clearly stained; (iii) a TS of a maize stem showing the scattered vascular bundles with xylem (large, thin-walled vessels stained red) and phloem (smaller, dense cells). Learners complete a structured observation table: Structure → What I see → Likely function (my best guess). They also make a scientific drawing of the root TS, labelling at least 4 structures. ACTIVITY B — Capillary Tube	ACTIVITY A: Distribute Handout 2A. Say: 'You have 20 minutes. Work in pairs. First, read the labels silently — 2 minutes. Then complete the observation table. Then make your scientific drawing. I will circulate. If you are unsure of a label, write your best guess and put a question mark.' Circulate; at 8 minutes, pause the class and cold-call one pair to describe what they see in the xylem vessels vs phloem cells: 'Amina and Brian — describe the xylem to us in your own words. What do you notice about the size and colour of these cells?' [WAIT TIME: 12 seconds.] After their response, add: 'Notice the xylem vessels are large and hollow — they are dead cells. We will come back to why that matters.' ACTIVITY B: Before starting, say: 'Watch carefully. I am using the	Structured Observation with Scaffolded Recording (Observation Table + Scientific Drawing) — the three-column table (Structure / What I see / Likely function) separates raw observation from interpretation, preventing premature closure. The scientific drawing forces learners to attend to spatial relationships (e.g., xylem position relative to phloem) rather than passively viewing. The capillary demonstration provides a physical analogue for the abstract concept of capillary rise, making the invisible mechanism in xylem vessels visible and measurable. The gallery walk with sticky-note questions externalises thinking and generates authentic questions for the Driving Question Board.	Teacher checks observation tables during circulation — specific indicator: learners should correctly identify xylem as large, hollow, and red-stained (or otherwise prominent) in at least 2 of the 3 diagrams. Teacher checks scientific drawings for presence of at least 4 correct labels. During the capillary demo, teacher listens for the key observation: 'the narrowest tube has the highest water.' Sticky notes from the gallery walk are collected and sorted — teacher uses these to identify the 2–3 most prevalent misconceptions to address in the Explain phase (Lesson 3).

	Demonstration (15 min): Teacher performs a live demonstration at the front. Three glass capillary tubes of different diameters (wide ~3 mm, medium ~1 mm, narrow ~0.3 mm) are held vertically in a beaker of water coloured red with food dye — the same dye used in the phenomenon. Learners observe and record: (a) the height water rises in each tube; (b) the shape of the meniscus; (c) which tube's water rises highest. Learners record observations on Handout 2B. ACTIVITY C — Gallery Walk (10 min): Six A2 diagrams are posted around the room: (1) Root hair cell 3D diagram; (2) Root TS with xylem/phloem labelled; (3) Stem TS of maize; (4) Stem TS of bean; (5) Diagram of cohesion (water molecule chain); (6) Diagram of adhesion (water clinging to glass/cell wall). Pairs spend ~90 seconds at each station, writing one observation and one question on a sticky note left at each poster.	same red food dye we placed the maize seedling roots in. What do you predict will happen — write one sentence NOW.' [30-second writing pause.] Slowly lower all three tubes simultaneously into the beaker. Say nothing for 45 seconds — allow learners to observe in silence. Then ask: 'What do you notice? Tell your partner — 30 seconds.' Cold-call 2 pairs. Write their exact words on the board. Then ask: 'Which tube reminds you most of the xylem vessels you just drew?' [WAIT TIME: 15 seconds.] Cold-call 1 learner. ACTIVITY C: Assign roles — one partner is the 'Recorder', the other is the 'Reader'. Set a timer. Say: 'One observation, one question — sticky note at each poster. Move when the timer beeps. Go.' Circulate and read sticky notes; mentally note any misconceptions (e.g., confusing xylem with phloem, or thinking water is 'pumped' by the roots like a heart).		
Explain Phase  VIDEO: Intro to exponential functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Geometric Visual of a Square Root (PLIX) Source: CK-12  Search ARES for similar readings	Learners participate in a whole-class 'Connect the Evidence' discussion. The teacher uses a split board: on the left, the phenomenon image (red-stained maize stem TS); on the right, the evidence gathered today. Learners are asked to draw explicit connections: 'Which structure in your photomicrograph matches the red-stained ring in the phenomenon?' and 'How does the capillary demonstration help explain how water gets to the top of a 2-metre maize plant in a Rift Valley farm?' Learners then write a 3-sentence 'evidence statement' in	Draw a T-chart on the board: left column 'Phenomenon — what we saw', right column 'Today's evidence — what it tells us'. Say: 'Let's build this together. What did we observe in the phenomenon? Chiamaka?' [WAIT TIME: 10 seconds.] Fill in the left column with learner responses. Then: 'Now — which piece of today's evidence connects to that observation? Look at your handouts. You have 2 minutes to find one connection and write it in your journal.' [Allow 2 minutes.] Cold-call 3 learners (aim for gender balance — use name	Evidence-to-Phenomenon Mapping (T-Chart) — by explicitly mapping each piece of new evidence onto the anchoring phenomenon, learners construct explanatory links rather than memorising isolated facts. The 3-sentence journal entry ('I observed... This suggests... I am still unsure about...') is a metacognitive writing-to-learn strategy that consolidates understanding and surfaces remaining gaps — feeding directly into the Driving Question Board.	Teacher reads 5–6 journal entries during the DQB phase (below) while learners are writing on the board. Specific indicator: the evidence statement should name at least one structure (xylem or root hair cell) AND at least one force (adhesion or cohesion) AND connect both to the phenomenon. If fewer than half the class achieves this, teacher opens Lesson 3 with a targeted re-teach of the structural–mechanical link.

	their science journals: 'Today I observed... This evidence suggests... One thing I am still unsure about is...'	sticks). After each, say: 'Does anyone want to add to or challenge that connection?' [WAIT TIME: 12 seconds.] Conclude by asking: 'So — the narrow capillary tube and the narrow xylem vessel — what do they have in common?' Cold-call 1 learner. Then say: 'We have a name for the forces that make water climb: adhesion — water sticking to the wall — and cohesion — water molecules sticking to each other. Write those two words and their definitions in your journal now.' [Allow 90 seconds.]		
Driving Question Board (DQB) Creation  VIDEO: Intro to exponential functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Geometric Visual of a Square Root (PLIX) Source: CK-12  Search ARES for similar readings	Each learner receives two sticky notes. On the green note, they write one thing today's evidence has helped them understand about the driving question: 'How does a maize plant move water from soil to its topmost leaves?' On the orange note, they write one new question today's evidence has raised. Learners post both notes on the class DQB (a large sheet of paper divided into: 'What we now have evidence for' / 'Questions we still need to answer'). The teacher reads 5–6 notes aloud, groups similar questions, and explicitly connects them to the upcoming lessons in the sequence (e.g., 'Three of you asked about what drives water ALL the way up — that is exactly what Lesson 3 is about').	Distribute sticky notes. Say: 'Green note — one thing today's evidence helped you understand. Orange note — one new question it raised. You have 3 minutes. Write in full sentences.' [Allow 3 minutes.] Invite learners to post notes on the DQB. Read 5–6 aloud (read the question, do NOT answer it yet). Group similar questions with a bracket. Say: 'Look at these orange questions — many of you are asking HOW exactly water gets pulled all the way to the top of a tall maize stalk. That is the cohesion-tension theory, and it is the focus of Lesson 3. Hold those questions — they are exactly the right ones.' Point to the driving question posted at the top of the DQB and say: 'We have not fully answered this yet — but today's evidence has moved us forward. What part can we now answer with confidence?' Cold-call 2 learners. [WAIT TIME: 15 seconds each.]	Two-Colour Sticky-Note DQB — separating 'evidence-based understanding' (green) from 'new questions' (orange) makes the distinction between claims and evidence explicit, and models scientific practice. Reading questions aloud without answering them signals that questions are intellectually valued, motivating continued inquiry. Grouping similar questions demonstrates the power of collective sensemaking and previews the lesson sequence.	Teacher reads all posted sticky notes after class. Specific indicators: green notes should reference at least one observable structure or demonstrated force; orange notes should be genuine mechanistic questions (e.g., 'What pulls the water upward?') rather than factual recall questions (e.g., 'What is xylem?'). A prevalence of factual questions signals that learners have not yet moved from description to causal reasoning — teacher adjusts Lesson 3 opening accordingly.
Model Building Phase  VIDEO:	Learners retrieve their Lesson 1 initial model (a sketch showing how they thought water moved from soil to leaf). Using a different	Say: 'Get out your Lesson 1 model. Pick up a different colour pen — any colour that is NOT the one you used in Lesson 1. You are	Cumulative Model Revision with Colour-Coded Layering — using a different colour pen creates a visible record of how	Teacher collects or photographs 6–8 models. Specific indicators: (1) Root hair cells drawn at root surface — not inside root; (2)

Intro to exponential functions Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Geometric Visual of a Square Root (PLIX) Source: CK-12 Search ARES for similar readings	colour pen, they annotate or redraw their model to incorporate today's evidence, adding: (a) root hair cells at the soil-root interface; (b) xylem vessels in the stem (narrow tubes in the correct position relative to phloem); (c) an arrow labelled 'adhesion' (water-to-wall) and 'cohesion' (water-to-water) inside the xylem. Learners do NOT need to be correct about the mechanism yet — the goal is to update the model with structural evidence. They write one sentence below the model: 'My model changed because today I saw evidence that...'	going to update your model with today's evidence. You have 7 minutes. Add at least three new things: root hair cells, xylem vessels, and the labels adhesion and cohesion. You do not need to know exactly how it works yet — just add what you have observed.' [Set timer for 7 minutes. Circulate. Coach 3–4 learners individually: 'Where in your model would you place the xylem? Think back to the cross-section diagram.' At 5 minutes, say: 'One minute left — write your sentence at the bottom: My model changed because today I saw evidence that...' At the close, ask 2 learners to hold up their revised model and describe one change to the class. Affirm: 'Notice how Wanjiru has drawn the xylem as a narrow tube — that connects directly to what we saw in the capillary demonstration.'	understanding evolves across lessons, making the learning progression tangible for both learner and teacher. The sentence stem 'My model changed because today I saw evidence that...' enforces evidence-based reasoning rather than copying from a textbook. Models will be revisited and further annotated in Lessons 3–8, ultimately becoming a complete mechanistic model of water and nutrient transport in a maize plant.	Xylem drawn as hollow/tube-like structures in the stem; (3) At least one force label (adhesion or cohesion) present. These models serve as a pre-assessment baseline for Lesson 3's mechanistic content. Models with xylem absent or phloem/xylem confused will receive targeted feedback at the start of Lesson 3.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners correctly identify xylem (not phloem) as the tissue stained red in the phenomenon? If many confused the two, prepare a 5-minute side-by-side comparison at the start of Lesson 3. (2) Did the capillary demonstration generate genuine 'aha' moments, or did learners watch it passively? If passive, in future lessons add a prediction step with individual mini-whiteboards before the demo begins. (3) Were the orange DQB questions mechanistic (asking 'how' or 'why') or merely factual ('what is...')? If mainly factual, Lesson 3 should begin with a structured 'Notice and Wonder' routine rather than jumping directly to the cohesion-tension theory. (4) Were learners from all groups (including those who struggle with English scientific vocabulary) able to access the photomicrographs? If not, consider pairing Swahili-language captions with the diagrams in the handout for the next lesson. (5) Did you manage time effectively across the three activities in Phase 2? The capillary demonstration is the most time-sensitive — if it ran short, prioritise it first in a future iteration.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Photomicrographs of root hair cells and transverse sections of root and stem showing xylem and phloem; a live capillary-tube demonstration using red food dye showing water rising higher in narrower tubes; six gallery-walk diagrams showing cell structures and molecular forces.
What did I learn?	Root hair cells have a large surface area that maximises water absorption from the soil. Xylem vessels are narrow, hollow, dead cells arranged in rings or bundles in the stem — these are the channels the red dye travels through in the phenomenon. Phloem is distinct from xylem and carries dissolved sugars. Adhesion (water sticking to xylem walls) and cohesion (water molecules sticking

	to each other) are the physical forces that help water rise in narrow tubes — just as water rises higher in a narrower capillary tube.
How does this explain the phenomenon?	The red-stained rings in the cross-sectionally cut maize stem from Kisumu are xylem vessels — not phloem or any other tissue. Water is able to enter these narrow vessels partly because of adhesive and cohesive forces, analogous to the capillary rise seen in the demonstration. Root hair cells increase the surface area available for water absorption from the soil. These structural observations begin to explain how water moves upward against gravity, though the full mechanism (cohesion-tension theory, transpiration pull) will be established in Lesson 3.

LESSON 3 (80 minutes): Absorption of Water and Mineral Salts: Osmosis, Active Transport, and Root Pressure

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To explain how water and mineral salts are absorbed by root hair cells and transported upward through the plant using osmosis, active transport, and root pressure mechanisms.
Knowledge	Learners describe absorption of water and mineral salts in plants (KICD SLO a), explaining the roles of osmosis, active transport, and root pressure in moving substances from the soil into and up the root.
Skills	Learners use print and/or digital media to research how mineral salts such as DAP (Di-ammonium Phosphate) enter root hair cells; they construct written explanations linking evidence from the red-dye phenomenon to the mechanisms discovered [SEP: Constructing Explanations].
Attitudes	Learners appreciate the role of transport in plants (KICD SLO e) and the importance of responsible fertiliser use — including DAP, CAN, and Urea — in sustaining maize and sukuma wiki production on Kenyan farms.
Key Inquiry Question	Why is transport in plants important?
Purpose in Storyline	In Lesson 1 learners observed that red dye travels upward through specific woody channels in the maize stem, and in Lesson 2 they gathered evidence on xylem and phloem structure. Lesson 3 now asks: HOW does water first get INTO the root and then get PUSHED up into those channels? Learners construct explanations for the entry of water (osmosis), the entry of mineral salts (active transport), and the initial upward push (root pressure) — directly answering why the dye moved up rather than staying at the bottom of the beaker.
Safety Notes	No hazardous chemicals are used in this lesson. If printed diagrams are laminated, handle laminator with care. Remind learners not to ingest any food dye or salt solutions used for demonstration purposes. Wash hands after handling soil or plant specimens brought in from the school garden.

B. LESSON OVERVIEW

Lesson 3 is the EXPLAIN phase of the evidence-gathering sequence. Building on the red-dye and polythene-bag phenomena from Lessons 1 and 2, learners now construct scientific explanations for HOW water and mineral salts enter a maize root and begin their upward journey. Through a structured Predict-first discussion, a guided jigsaw research task using print/digital media, and a whole-class sensemaking discussion, learners distinguish between osmosis (water entry via a concentration gradient across the semi-permeable root hair cell membrane), active transport (energy-requiring uptake of mineral ions such as phosphate from DAP fertiliser), and root pressure (the resulting hydrostatic push that drives water up into the xylem). Kenyan agricultural context — fertiliser use on maize and sukuma wiki farms in Rift Valley, Kisumu, and Central Kenya — is woven throughout. Learners close by updating the Driving Question Board and revising their cumulative transport model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Learners recall the red-dye observation from Lesson 1: the dye moved upward through the	Display the two jars and the root hair cell diagram on the board or projector. Say: 'Look carefully at	Think-Pair-Share Prediction Wall. By externalising individual predictions before instruction,	Observable indicator: Every learner has produced a written 2-sentence prediction in their

Osmosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12 Search ARES for similar readings	maize stem against gravity. The teacher displays a large printed photograph of a maize root hair cell diagram alongside a jar of soil-water (murky water with dissolved salts) and a jar of plain water. Learners individually write a 2-sentence prediction in their science notebooks answering: 'The water inside a root hair cell is less/more concentrated than the water in the soil — therefore water will move IN/OUT of the cell.' They then share predictions with a shoulder partner (Think-Pair) before three cold-calls to the class. Learners also predict whether a maize plant fertilised with DAP near Kericho would absorb more or fewer mineral ions than an unfertilised plant, and why.	these two solutions. Before I tell you anything new today, I want you to think about what you already believe.' Pause for 30 seconds of silent thinking. Then instruct: 'Write your prediction — two sentences — NOW. You have 90 seconds.' After writing, say: 'Turn to your partner. You have 60 seconds each.' After pair-share, cold-call exactly THREE learners using name cards or a random selector. For each response, ask one follow-up probe: 'What evidence from our Lesson 1 observation makes you say that?' Allow 10–12 seconds WAIT TIME after each cold-call before accepting an answer. Record contrasting predictions (water moves IN vs. water moves OUT) in two columns on the board labelled PREDICT — leave these visible throughout the lesson for revisiting. Do NOT correct predictions at this stage.	learners activate prior knowledge of concentration and diffusion from Junior School Integrated Science. The visible two-column Prediction Wall creates cognitive dissonance that motivates the research task and gives the teacher real-time diagnostic data on misconceptions (e.g., learners who predict water moves OUT of the root).	notebook within the 90-second window. The teacher scans notebooks during the pair-share, noting (a) learners who correctly predict inward osmosis, (b) learners who confuse direction, and (c) learners who cannot explain using concentration. These three groups inform targeted questioning during the Explain phase. Target: ≥80% of learners produce a written prediction that uses the word 'concentrated' or 'concentration' even if the direction is wrong.
Observe Phase  VIDEO: Osmosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12 Search ARES for similar readings	Learners work in groups of four. Each group receives one laminated ARES resource card (or accesses the ARES offline digital module) covering one of four topics: (A) Osmosis and Water Potential in Root Hair Cells, (B) Active Transport of Mineral Ions — Focus on Phosphate from DAP, (C) Root Pressure and the Hydrostatic Push into Xylem, (D) Consequences of Fertiliser Overuse — Plasmolysis and Wilting in Sukuma Wiki. Groups read/view their resource for 8 minutes, then each group member becomes an 'expert' who will teach the other three. After 3 minutes of individual note-taking on a structured Evidence Record sheet (three columns: What I	Before releasing groups, say: 'Your job is not just to read — your job is to gather EVIDENCE that will help you explain why the red dye moved up through those wooden rings in our maize stem. Every fact you find must connect back to that stem.' Distribute or open resource cards. Set a visible countdown timer for 8 minutes (reading) + 3 minutes (individual notes) + 8 minutes (jigsaw teaching). Circulate during jigsaw teaching. Stop at Group B discussions and ask: 'When a Rift Valley farmer adds DAP fertiliser to the soil, what happens to the concentration of phosphate ions around the root — does it go UP or DOWN?' Wait 10 seconds. Cold-call one learner from the group.	Expert Jigsaw with Anchored Evidence Records. The jigsaw structure ensures every learner encounters all four topics while developing accountability as a peer teacher. The Evidence Record sheet's third column ('Connection to our maize phenomenon') forces learners to continuously tether new information to the anchoring phenomenon rather than treating it as disconnected facts. This is a core NGSS SEP: Obtaining, Evaluating, and Communicating Information scaffold.	Observable indicator: Each learner's Evidence Record sheet has at least one entry in all three columns for each of the four topics by the end of the jigsaw. The teacher spot-checks five to six sheets during circulation, specifically checking Column 3 (connection to phenomenon). Learners who leave Column 3 blank are prompted: 'How does this help you explain what happened in our maize stem on Day 1?' Additionally, the teacher listens for accuracy during jigsaw teaching — if a Group B expert incorrectly describes active transport as passive, the teacher intervenes with a Socratic prompt: 'Does this process require the plant to spend energy — yes or

	observed/read What it means Connection to our maize phenomenon), jigsaw groups of four (one expert per topic) form and each expert teaches their topic in 2 minutes. All learners complete their Evidence Record sheet across all four topics. The teacher circulates and listens to jigsaw teaching for accuracy.	Stop at Group D discussions and ask: 'A sukuma wiki farmer in Kiambu over-applies CAN fertiliser. She notices the leaves wilting the next morning even though she watered the night before. What does your evidence card say about why this might happen?' Wait 12 seconds before accepting an answer. After jigsaw teaching, bring the class back together. Write the three key terms on the board: OSMOSIS ACTIVE TRANSPORT ROOT PRESSURE. Say: 'In the next phase, you are going to use these three ideas to explain our phenomenon — so hold onto your Evidence Record sheets.'		no? Where does that energy come from?
Explain Phase  VIDEO: Osmosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12  Search ARES for similar readings	Whole class reconvenes. The teacher uses a large annotated diagram of a maize root cross-section (projected or drawn on the board) showing: soil water, root hair cell with vacuole, cell membrane, cortex cells, endodermis (Casparian strip), and xylem vessel. Working in their original groups, learners construct a written Claim-Evidence-Reasoning (CER) explanation answering: 'How does water from the soil in our Kisumu school garden reach the xylem vessel of a maize plant — and how does it then get pushed upward?' Each group writes on a shared A3 sheet. Groups then share key sentences with the class in a gallery-walk format (sheets posted on wall, learners rotate for 3 minutes, adding sticky-note agreements or challenges). The teacher leads a 10-minute whole-class discussion synthesising the three mechanisms, correcting misconceptions, and explicitly	Introduce CER explicitly: 'A scientific explanation has three parts. Claim — your answer. Evidence — data or observations that support it. Reasoning — the science principle that links them.' Write CER on the board with a maize-farm example: 'Claim: Water enters the root hair cell from the soil. Evidence: In Lesson 1, the red dye moved upward. Reasoning: Osmosis — water moves from high water potential (soil) to low water potential (root cell vacuole) across the semi-permeable membrane.' Give groups exactly 7 minutes to write their CER on A3 paper. During gallery walk, prompt readers with: 'What is the EVIDENCE in this group's explanation? Is it from our observations or from the resource cards?' After gallery walk, lead synthesis: point to the root diagram and trace the journey — soil → root hair → cortex → xylem — narrating each mechanism aloud. At the xylem entry point,	Claim-Evidence-Reasoning (CER) Framework + Gallery Walk. CER structures scientific writing so learners do not simply restate facts but explicitly justify them using evidence. The gallery walk introduces peer critique as a sensemaking tool — learners must evaluate whether another group's evidence is valid (from observations) or merely asserted. The public revisit of the Prediction Wall closes the predict-observe-explain loop, making conceptual change visible to learners themselves.	Observable indicator: Each group's CER sheet contains (a) an explicit claim naming at least two of the three mechanisms (osmosis, active transport, root pressure), (b) at least one piece of evidence traceable to either the Lesson 1 phenomenon or the Lesson 2/3 research, and (c) a reasoning statement that uses a scientific principle (e.g., concentration gradient, semi-permeable membrane, ATP). The teacher collects all A3 sheets. Any CER that lacks a reasoning statement is flagged for targeted feedback before Lesson 4. Additionally, the self-stick dot activity gives the teacher a rapid visual of how many learners (target: ≥75%) correctly self-identify the inward direction of osmosis after instruction.

	connecting each mechanism back to the red-dye observation. Specific connection: 'Root pressure is the pump — it is why the red dye we saw moved UP into the stem, not just sideways.' Learners also discuss why DAP (Di-ammonium Phosphate) fertiliser, widely used on maize farms in Uasin Gishu and Trans-Nzoia, improves absorption — because active transport carriers in root hair cells take up phosphate and ammonium ions, improving overall water potential gradients.	say: 'This build-up of water in the xylem creates pressure — ROOT PRESSURE. This is the first engine that pushes water up. Does this help us explain our red dye going UP against gravity?' Cold-call TWO learners. Wait 12 seconds each. Then ask: 'Why would a maize farmer in Eldoret who never adds DAP see slower water transport than one who does?' Cold-call ONE learner. Accept answers, then clarify: phosphate ions enter via active transport, increasing solute concentration in root cells, steepening the osmotic gradient and enhancing water uptake. Explicitly revisit the Prediction Wall: 'Look at our predictions from earlier. Who had the right direction for osmosis — and who needs to revise?' Ask learners to place a self-stick dot on the correct prediction column.		
Driving Question Board (DQB) Creation  VIDEO: Osmosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12  Search ARES for similar readings	The teacher displays the class Driving Question Board, which carries the anchoring question and sticky notes from Lessons 1 and 2. Learners individually write NEW sticky notes (green = new explanations, orange = new questions that arose today). Each learner posts at least one green note and one orange note. Sample green notes expected: 'Water enters root hair cells by osmosis because the cell sap is more concentrated than soil water'; 'DAP fertiliser adds phosphate ions that are pulled in by active transport using ATP'; 'Root pressure pushes water up into the xylem — that is why the dye rose.' Sample orange notes expected: 'If root pressure starts the push, what keeps the water going up to the topmost	Hand each learner two sticky notes (one green, one orange — or use different coloured pens on the same colour note). Say: 'Green notes are for things we can NOW explain. Orange notes are for new questions your brain is asking — things today's lesson did NOT fully answer.' Allow 3 minutes of silent individual writing. Then say: 'Walk up, post your notes, and read two others before you sit down.' After posting, stand at the DQB and narrate aloud: 'Look — many of us are now asking: root pressure explains the START of the journey, but a maize plant in Kisumu grows two metres tall. Is root pressure strong enough to push water all the way to the top leaf? That is exactly where Lesson 4 takes us.' Point to any orange notes about	Two-Colour DQB Protocol (Explain vs. Question). Separating new explanations (green) from new questions (orange) makes the asymmetry between current understanding and remaining unknowns visible on a single board. This metacognitive tool helps learners see that science is cumulative — each lesson resolves some questions and opens new ones — and it creates authentic motivation for the next lesson in the sequence.	Observable indicator: Every learner posts at least one green note containing a mechanistic explanation (not merely a label). The teacher scans green notes for naïve explanations such as 'water goes in because plants need water' — these lack mechanism and are earmarked for follow-up. Orange notes that identify the cohesion-tension gap ('root pressure is not enough for tall plants') indicate higher-order thinking and can be used to accelerate instruction in Lesson 4. Collect a photo of the DQB for the ARES platform record.

	maize leaf — is root pressure enough?'; 'Why does the dye only go into the xylem rings and not the phloem?'; 'What happens if the soil becomes too salty from too much fertiliser?' The class briefly reviews the board together: the teacher groups related notes and identifies the most important unresolved question, which becomes the bridge to Lesson 4 (translocation and the cohesion-tension mechanism).	'too much fertiliser' and say: 'This connects to what we said about CAN and DAP — over-applying fertiliser reverses the osmotic gradient. We called that PLASMOLYSIS. Remember that word — it will appear again in Plant Growth.' Do NOT answer the Lesson 4 questions now — preserve genuine inquiry tension.		
Model Building Phase  VIDEO: Osmosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12  Search ARES for similar readings	Learners retrieve their transport model diagram from Lesson 2 (a sketch of the maize plant showing the xylem pathway). They now REVISE and ANNOTATE it to add the root-level mechanisms explained today. Required additions: (1) Root hair cell with arrows showing water entering by osmosis (labelled with 'high water potential → low water potential'); (2) Active transport carriers on the root hair cell membrane with ATP lightning bolt symbols, showing mineral ion (phosphate/NH_4^+) entry; (3) A ROOT PRESSURE arrow inside the xylem pointing upward; (4) A soil annotation noting 'DAP fertiliser increases phosphate concentration in soil — active transport required.' Learners also write a one-sentence MODEL LIMITATION note at the bottom: 'This model does not yet explain how water continues rising above the root — that is the next question.' Revised models are initialised by a peer before being stored in the ARES portfolio.	Distribute or ask learners to open their Lesson 2 model sketches. Say: 'Your model from Lesson 2 showed WHERE water travels in the plant. Today you are adding HOW it gets in at the root level. A better model explains mechanism, not just structure.' Project a sample annotated root hair cell on the board (without the labels filled in) as a scaffold. Say: 'You have 8 minutes. Add at least FOUR new annotations to your model — I will come around and check.' Circulate and prompt: 'Show me where osmosis is happening on your diagram.' 'Where is energy being used? Mark it.' 'If a farmer adds too much urea near these roots, what would happen to your water potential arrows — would they reverse?' Allow a 2-minute peer-initialling swap at the end. Collect five to six models for formative review before Lesson 4. Close the lesson by asking the class: 'Our model now explains the root. But look at this maize plant photo — the water still has to climb two metres. What is pulling it up? That is our next investigation.'	Iterative Model Annotation (Progressive Model Building). Requiring learners to annotate rather than redraw forces them to reconcile new mechanistic knowledge with their existing structural model, promoting conceptual integration. The 'Model Limitation' sentence is a deliberate metacognitive move — it teaches learners that a good scientific model explicitly acknowledges what it cannot yet explain, mirroring authentic scientific practice (NGSS SEP: Developing and Using Models). The peer initialling introduces low-stakes peer assessment.	Observable indicator: Revised models contain all four required additions. The teacher specifically checks that osmosis arrows are drawn in the correct direction (soil → root hair cell) and that active transport is distinguished from osmosis by the presence of an energy marker (ATP symbol or the word 'energy required'). Models where osmosis arrows are reversed or where active transport is labelled as diffusion are flagged. During Lesson 4 warm-up, the teacher returns two or three anonymised flagged models to the class for whole-group revision as a re-teaching moment.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your ARES teacher journal:

1. **PREDICTION WALL** — How many learners initially predicted the wrong direction for osmosis (water OUT of the root)? Did the lesson successfully shift this misconception? What evidence (sticky dots, CER sheets, model arrows) confirms the shift?
2. **JIGSAW QUALITY** — Were learners able to accurately teach their assigned topic to peers, or did certain topics (e.g., active transport and ATP; root pressure mechanism) produce inaccurate peer-teaching? Which resource cards may need to be simplified or supplemented with a short teacher-led mini-explanation before Lesson 4?
3. **KENYA CONTEXT INTEGRATION** — Did learners make genuine connections between DAP/CAN fertiliser use and the mechanisms of mineral salt absorption, or did the agricultural context feel grafted on? Consider whether a brief visit to the school's own kitchen garden (if applicable) or a photo gallery of Trans-Nzoia maize farms would deepen engagement in Lesson 4.
4. **CER DEPTH** — Scan the A3 CER sheets collected. Were reasoning statements mechanistic (referencing concentration gradients, semi-permeable membranes, ATP) or surface-level ('because plants need water')? Plan a 5-minute CER quality warm-up at the start of Lesson 4 using an anonymised strong example and an anonymised weak example.
5. **DQB ORANGE NOTES** — Count how many orange notes correctly identify the cohesion-tension gap (root pressure alone insufficient for a 2-metre maize plant). This is the conceptual readiness indicator for Lesson 4. If fewer than half the class produced this question organically, plan an explicit hook at the start of Lesson 4 — e.g., 'Root pressure generates about 0.1–0.2 MPa. A 2-metre maize plant needs approximately 0.2 MPa just to overcome gravity. So we are right at the limit — what else must be happening?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In Lesson 1 we observed that red food dye placed in a beaker of water entered the roots of a freshly uprooted maize seedling from our Kisumu school garden and moved upward, appearing as vivid red rings only inside the xylem tissue when the stem was cut cross-sectionally. In Lesson 2 we identified the xylem and phloem as the transport channels. In today's lesson we examined diagrams and research resources showing root hair cells in contact with soil water, and we observed that the cell sap inside the root hair cell vacuole is more concentrated (lower water potential) than the water in the surrounding soil.
What did I learn?	Water enters root hair cells by OSMOSIS — it moves from a region of higher water potential (dilute soil water) to a region of lower water potential (concentrated cell sap) across the semi-permeable cell membrane. Mineral salts such as phosphate ions from DAP fertiliser and ammonium ions from CAN or Urea cannot enter by osmosis; they are absorbed by ACTIVE TRANSPORT — protein carrier molecules in the cell membrane use ATP energy to pump ions against their concentration gradient into the root hair cell. As water and mineral ions accumulate in the root cells and pass inward through the cortex to the xylem, they create a hydrostatic pressure called ROOT PRESSURE , which pushes water upward into and through the xylem vessels. This is why the red dye in our phenomenon moved upward against gravity. Over-application of fertiliser increases solute concentration in the soil, reversing the water potential gradient and causing water to leave the root cells (plasmolysis), which is why sukuma wiki or maize can wilt after excess fertiliser is applied.
How does this explain the phenomenon?	The red dye entered the maize root and rose through the xylem because: (1) osmosis drove water molecules from the relatively dilute soil water into the more concentrated root hair cell vacuole across the semi-permeable membrane; (2) active transport used ATP to pump mineral ions such as phosphate (from DAP) into the root cells against their concentration gradient, further lowering water potential and sustaining the osmotic gradient; and (3) the accumulation of water in the xylem generated root pressure — a hydrostatic push that propelled the dye-coloured water upward through the xylem channels, which is why only the xylem rings (not the phloem or pith) turned red. Our model now explains the ROOT END of the journey. However, root pressure alone may not be sufficient to lift water to the topmost leaves of a two-metre maize plant — this is the unresolved question we carry into Lesson 4.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Translocation: How Does the Maize Plant Move Sugar from Leaf to Root?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners will explain the process of translocation in plants by analysing evidence from a ring-barking (girdling) demonstration and comparing xylem and phloem function, connecting their findings to the anchoring phenomenon of how the Kisumu maize plant moves substances from source to sink.
Knowledge	Learners explain translocation in plants, describing the source-to-sink model of phloem transport and distinguishing between the roles of xylem (water and mineral salts, upward) and phloem (sugars and assimilates, bidirectional, source to sink).
Skills	Learners analyse and interpret data from a ring-barking/girdling demonstration; construct explanations for why the coloured-water dye in the anchoring phenomenon moves only through xylem tissue; complete a two-column xylem-vs-phloem comparison chart; and update the Driving Question Board with translocation evidence. [SEP: Analysing and Interpreting Data; Constructing Explanations]
Attitudes	Learners appreciate the role of transport in plants — including translocation — in sustaining plant life, crop productivity, and food security for Kenyan farmers in regions such as Kisumu, Kericho, and the Rift Valley.
Key Inquiry Question	Why is transport in plants important?
Purpose in Storyline	In Lessons 1–3 learners established that water and mineral salts move upward through xylem vessels (demonstrated by the red-dye maize experiment). Lesson 4 now asks: what about the sugars made by photosynthesis in the leaves — how do they reach the roots, the developing maize cob, or the growing tip? By observing the girdling demonstration and analysing phloem exudate evidence, learners discover the second transport tissue — phloem — and the source-to-sink model. This advances the driving question by explaining not only HOW water moves up, but HOW manufactured food moves to where it is needed throughout the plant.
Safety Notes	Razor blades or scalpels used in the ring-barking demonstration must be handled only by the teacher or a designated learner under direct supervision. Freshly cut stem surfaces may be slippery — place specimens in a tray. Learners must wash hands after handling plant material. No open wounds near cut plant surfaces. If using a bean or sweet-potato stem, avoid touching eyes after handling. All sharp instruments to be collected and counted at the end of the lesson.

B. LESSON OVERVIEW

Learners begin by revisiting their Driving Question Board and predicting what tissue carries sugar from a maize leaf down to the roots and developing cob. The teacher then leads a ring-barking (girdling) demonstration on a soft-stemmed plant (e.g., a bean or sweet-potato cutting from the school garden), directing learners to observe the swelling of tissue just above the ring cut over time (or via pre-prepared specimens showing the swelling). Learners also examine photographic/diagrammatic evidence of phloem exudate from aphid stylet experiments and sieve-tube diagrams. Using an Analysing-and-Interpreting-Data routine, pairs record observations in a structured data table, then use a two-column Xylem-vs-Phloem chart to construct an explanation. A source-to-sink model is introduced through a worked example using a maize plant (leaves as source, roots and cob as sinks). Learners then construct written explanations linking both the red-dye xylem evidence (Lessons 1–3) and the new phloem/girdling evidence to the driving question. The DQB is updated collectively. The lesson closes with a teacher-facilitated sensemaking discussion tying translocation to the real-world importance of watering, manuring, and fertilising crops on Kenyan farms.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 2D divergence theorem Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: How Scientific Teams...:Translocation Activity after Segment 3 (PDF Format) Source: MIT Blossoms  Search ARES for similar readings	Learners individually write a prediction (2–3 sentences) in their science journals answering: 'In a maize plant growing on a farm near Kisumu, the leaves make sugar by photosynthesis. Predict: how does that sugar reach the roots and the developing maize cob? Does it travel through the same red-stained channels we saw in Lesson 2, or through different channels — and why do you think so?' Learners share predictions with a shoulder partner (Think-Pair) for 2 minutes, then three learners are cold-called to share aloud.	Display the anchoring phenomenon image (cross-section of maize stem showing red-stained xylem rings) on the board alongside a diagram of a maize plant with arrows labelled 'sugar made here' at the leaves and question marks at the roots and cob. Say: 'In our last lessons we tracked water moving UP through those red rings. Today we ask a different question — what about the food the plant makes? Where does it go, and how?' Distribute science journals. Give learners 3 minutes of individual silent writing time. After Think-Pair, cold-call exactly 3 learners using name sticks: ask each 'What evidence from our previous lessons supports your prediction?' WAIT 12 seconds after each prompt before accepting an answer. Record key prediction terms (e.g., 'same tubes,' 'different tubes,' 'phloem,' 'flows down') on the board without evaluating correctness — label the board column 'Our Predictions (Lesson 4).'	Think-Pair-Predict: activates prior knowledge of xylem from Lessons 1–3 and creates cognitive dissonance — learners who predicted 'same tubes' will be productively challenged by the girdling evidence in the Observe phase. Recording predictions publicly ensures accountability and revisiting.	Teacher circulates during individual writing and notes (on a clipboard checklist) whether learners: (a) reference xylem by name, (b) propose a different tissue for sugar, or (c) give no tissue-specific answer. This identifies who needs targeted questioning during the Observe phase. Observable indicator: at least 70% of learners write a prediction that references a specific plant tissue.
Observe Phase  VIDEO: 2D divergence theorem Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: How Scientific Teams...:Translocation Activity after Segment 3 (PDF Format)	PART A — Ring-Barking Demonstration (15 min): The teacher (or designated learner under supervision) shows two pre-prepared bean or sweet-potato stem specimens: Specimen A — intact stem; Specimen B — a stem that was ring-barked (a 1–2 cm band of bark/phloem removed) 48 hours earlier, showing visible swelling/bulging of tissue just above the cut. Learners sketch both specimens in their journals, labelling the cut zone, the swelling above, and the region below. They	Before showing Specimen B, ask: 'Look carefully at the area just above the ring cut. What do you notice?' WAIT 12 seconds. Cold-call 2 learners. Then ask: 'If water was accumulating here, where would it have come from? But wait — water moves UP through xylem. So what else might be accumulating ABOVE the cut?' WAIT 15 seconds — this is the key productive-struggle moment; do not answer immediately. After learners propose ideas, say: 'You are thinking like scientists. Let us	Analysing and Interpreting Data (NGSS SEP 4): Learners move from raw observation (swelling above ring cut) to inference (phloem must carry something that cannot pass the cut) to pattern recognition (phloem moves substances from source to sink). The structured data table scaffolds this progression explicitly, preventing learners from jumping to conclusions without evidence.	Teacher checks completed Observation Data Tables during circulation — look for: (a) correct identification of swelling above (not below) the cut, (b) inference that phloem carries sugars/assimilates downward, (c) at least 3 of 5 rows correctly completed in the two-column chart. Cold-call one pair per Evidence Card to share their 'What this tells me' column. Observable indicator: pairs correctly identify the direction of phloem transport (source → sink) and name sugars/assimilates as

Format Source: MIT Blossoms Search ARES for similar readings	record: 'What is different between specimen A and specimen B? Where is the swelling, and what might have accumulated there?' PART B — Phloem Evidence Cards (10 min): Learners work in pairs to analyse two Evidence Cards (printed A5 sheets prepared by the teacher): Card 1 shows a labelled diagram of a sieve tube and companion cell with arrows indicating direction of flow; Card 2 shows a photograph and description of aphid stylet experiments (aphid inserts stylet into phloem, sugary sap flows out — used as evidence that phloem carries sugars under pressure). Pairs complete a structured Observation Data Table: Column 1 — Observation (what I see/read); Column 2 — What this tells me about phloem. PART C — Xylem-vs-Phloem Two-Column Chart (5 min): Each learner independently fills in a partially completed two-column chart comparing xylem and phloem across five attributes: tissue type, contents transported, direction of movement, cell structure (living/dead), and driving force.	look at our Evidence Cards to test your ideas.' During pair work on Evidence Cards, circulate and ask probing questions: 'On Card 2, the aphid stylet goes INTO the stem — which layer does it reach? What does the sugary sap tell us about what phloem carries?' After pairs complete their data tables, project a blank two-column chart on the board and ask pairs to contribute one row each using a round-robin format. Validate each row with: 'Does everyone agree? Give me a thumbs up if your chart matches.' Correct any misconceptions about directionality — emphasise: 'Xylem is one-way — soil to leaf. Phloem is source-to-sink — wherever sugar is made TO wherever sugar is needed.'		phloem contents.
Explain Phase  VIDEO: 2D divergence theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos  PDF: How Scientific Teams...:Translocation Activity after Segment 3 (PDF Format)	PART A — Source-to-Sink Mapping (10 min): Learners receive a printed outline diagram of a maize plant (roots, stem, leaves, developing cob). Working individually, they label: (i) two SOURCES of sugar (mature leaves), (ii) three SINKS (roots, developing cob, young growing leaves at the apex), and (iii) draw arrows showing the direction of phloem transport from each source to each sink. They also draw a separate set of arrows showing xylem transport direction, using a	Introduce the source-to-sink model with a Kenyan farming analogy: 'Think of a maize farm in Kisumu. The leaves are like the posho mill — they produce the food (ugali flour = sugar). The roots and the cob are like the market stalls — they receive and use the food. The phloem is like the matatu route connecting the mill to the market.' Write the analogy on the board. Then say: 'Use this model to label your maize diagram. Remember: a SOURCE is wherever sugar is MADE or STORED and then	Constructing Explanations (NGSS SEP 6): Learners synthesise two independent lines of evidence (red-dye xylem experiment + girdling demonstration) into a single coherent explanation that advances the driving question. The Kenyan matatu/posho mill analogy provides a culturally grounded conceptual anchor that makes the abstract source-to-sink model concrete and memorable.	Collect written explanations (or photograph journal pages) as an exit artefact. Assess against three criteria: (1) correct use of 'translocation' to describe phloem transport, (2) correct directionality (source → sink), (3) explicit contrast with xylem transport. Observable indicator: at least 65% of learners produce written explanations meeting all three criteria. Peer feedback round gives teacher real-time data on class-wide understanding.

Source: MIT Blossoms Search ARES for similar readings	different colour. PART B — Written Explanation (10 min): Learners write a 4–6 sentence explanation in their journals connecting BOTH the red-dye xylem evidence (from Lessons 1–3) AND the girdling evidence (today) to the driving question: 'How does a maize plant move water and nutrients from the soil all the way to its topmost leaves — and what does it do with the food it makes?' They must use the terms: xylem, phloem, translocation, source, sink, sieve tube, companion cell, and cohesion-tension (revision from Lesson 3). PART C — Peer Plenary Share (5 min): Three learners (cold-called) read their explanations aloud. Class listens and gives one 'star' (something strong) and one 'step' (something to improve) using a structured peer-feedback protocol.	released; a SINK is wherever sugar is USED or STORED.' During individual writing, circulate and prompt struggling learners: 'You described xylem in Lesson 2 — can you now write one sentence that CONTRASTS phloem with xylem?' For the plenary, after each cold-called learner reads, ask the class: 'Did they include the word TRANSLOCATION? Did they explain the direction clearly?' WAIT 10 seconds. Correct any conflation of transpiration and translocation — say: 'Transpiration is water LEAVING the plant through stomata. Translocation is food MOVING inside the plant through phloem. Different process, different tissue, different purpose.'		
Driving Question Board (DQB) Creation  VIDEO: 2D divergence theorem Source: Khan Academy (English - US curriculum) Search ARES for similar videos  PDF: How Scientific Teams...Translocation Activity after Segment 3 (PDF Format) Source: MIT Blossoms Search ARES for similar readings	Learners return to the class Driving Question Board (a large poster or section of the board maintained across all 8 lessons). In pairs, they write one new sticky note each: a GREEN note stating one new thing they can NOW explain about the maize plant's transport system (using evidence from today's lesson), and an ORANGE note stating one question that TODAY's lesson raised but has not yet answered (e.g., 'What powers phloem transport — is it pressure or something else?' or 'Does the maize plant always move sugar downward, or can it go upward too?'). Pairs post their notes and one representative per pair reads their green note aloud to the class.	Point to the driving question displayed prominently: 'How does a maize plant move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?' Say: 'After four lessons, let us audit what we can now explain — and what remains mysterious.' Give pairs 4 minutes to write their sticky notes. As pairs post notes, group the green notes into two clusters on the board: 'XYLEM evidence' and 'PHLOEM evidence.' Say: 'Look — we now have TWO transport systems mapped. Our first three lessons gave us the xylem half of the story. Today gave us the phloem half.' Read two or three orange notes aloud and say: 'These are excellent scientific questions. We will return to some of them in	Driving Question Board Curation: making thinking visible across the 8-lesson sequence. Green/orange sticky-note protocol separates 'evidence we have' from 'questions still open,' mirroring how scientists distinguish established findings from ongoing inquiry. Clustering notes by tissue type makes the two-transport-system architecture of the plant visually explicit.	Teacher scans all posted sticky notes during the plenary. Observable indicators: (a) green notes name 'phloem' or 'translocation' specifically (not vague 'tubes'), (b) orange notes reflect genuine conceptual curiosity rather than repetition of already-answered questions. Teacher flags pairs whose green notes contain misconceptions (e.g., 'phloem carries water') for targeted follow-up at lesson start of Lesson 5.

		Lessons 5 and 6 when we investigate transpiration.' Link explicitly to the second part of the anchoring phenomenon: 'Remember the polythene bag with water droplets? The water vapour came from XYLEM water leaving through stomata — transpiration. Our next lesson will investigate that. But today's lesson explained what happens to the SUGAR — it travels through PHLOEM by translocation.' WAIT 10 seconds after posing: 'Can anyone now explain why ring-barking a tree in Kericho eventually kills it, even though its xylem is undamaged?' Cold-call 2 learners.		
Model Building Phase  VIDEO: 2D divergence theorem Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: How Scientific Teams...:Translocation Activity after Segment 3 (PDF Format) Source: MIT Blossoms  Search ARES for similar readings	Learners retrieve their cumulative plant transport model diagram begun in Lesson 1 (a cross-sectional and longitudinal diagram of a maize stem and plant). Today they add the following to their model: (i) colour-in or shade the phloem tissue in a contrasting colour to xylem on the stem cross-section; (ii) draw and label sieve tubes and companion cells in the phloem region; (iii) add directional arrows for translocation (source leaves → sinks: roots, cob, apex) using a different arrow style (dashed) from xylem arrows (solid); (iv) write a model key that defines SOURCE, SINK, TRANSLOCATION, and PHLOEM. Learners then write a one-sentence 'model limitation' at the bottom: one thing their diagram does NOT yet show about phloem transport (e.g., 'My model does not show the pressure mechanism that drives phloem flow').	Distribute learners' model folders/booklets. Say: 'Every lesson we make our model more complete. In Lesson 1 your model showed only the outside of the maize plant. In Lesson 2 you added xylem vessels and the red-dye pathway. In Lesson 3 you added cohesion-tension forces. Today you add the SECOND transport system — phloem — and show translocation.' Project a reference diagram of a maize stem cross-section showing both xylem and phloem positions (vascular bundle). Say: 'Phloem is on the OUTER edge of the vascular bundle; xylem is on the inner edge — notice this on your cross-section.' Circulate and check: (a) arrows for phloem are going in the correct directions (not always downward — remind learners that young leaves at the apex are also sinks so phloem can move upward there too); (b) sieve tubes are labelled. After 8 minutes, cold-call one learner to hold up and describe their updated model to	Cumulative Model Revision: each lesson's model update is additive and explicitly linked to new evidence, so learners experience scientific modelling as an iterative, evidence-driven process rather than a one-off diagram. The 'model limitation' sentence develops metacognitive awareness of the difference between a model and reality — a core scientific practice.	Teacher reviews 5–6 models during circulation (not all — use a rotating roster across lessons). Check: (a) phloem correctly positioned on vascular bundle cross-section, (b) translocation arrows correctly labelled with source/sink, (c) model limitation sentence is scientifically meaningful. Observable indicator: 70% of models reviewed show phloem arrows in at least two directions (to roots AND to apex/cob), demonstrating understanding that 'sink' is context-dependent, not always 'downward.'

		the class. Ask the class: 'What has this model added since Lesson 3? What is still missing?' WAIT 12 seconds. Close by saying: 'A good scientific model is honest about its limitations. Writing your model limitation is not a weakness — it is what scientists do. It tells us where our next investigation must go.'	
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PREDICT PHASE: Did learners' initial predictions show residual confusion between xylem and phloem from earlier lessons? If more than 40% of learners predicted 'same tubes,' consider opening Lesson 5 with a brief retrieval quiz on the two-tissue distinction before proceeding to transpiration.
2. OBSERVE PHASE: Was the swelling on the pre-prepared ring-barked specimen clearly visible? If the specimen preparation (48 hours in advance) was insufficient, consider using a photograph of ring-barking evidence from a mature tree in the school compound or a Kenyan tea bush (*Camellia sinensis*) from Kericho, which shows dramatic phloem girdling effects. Document any specimen preparation timeline issues for future delivery.
3. EXPLAIN PHASE: During the plenary share, did learners consistently conflate 'translocation' with 'transpiration'? If yes, create a dedicated vocabulary anchor card (posted in the classroom for the remainder of the unit) contrasting the two terms with a simple Kenyan-context mnemonic: 'TransLOcation = fLOW of food through phLoem; TransPIRation = Plant Is Releasing water vapour.'
4. DQB UPDATE: Review the orange (unanswered question) sticky notes overnight. Cluster them and use the most common themes to open Lesson 5. If many learners asked about the energy source for phloem transport, consider adding a 5-minute ATP/active-loading hook to the start of Lesson 5.
5. MODEL BUILDING: Were learners able to correctly orient phloem on the outer edge of the vascular bundle without further prompting? If not, bring in a fresh maize stem cross-section in Lesson 5 for learners to observe directly under a hand lens, reinforcing the structural claim from the model.
6. DIFFERENTIATION NOTE: Learners who completed the two-column chart quickly and accurately can be challenged with an extension question: 'A sugarcane farmer in Kisumu cuts through the stem at harvest time and notices a sweet liquid oozing from the cut. Which tissue is this liquid coming from, and how does the source-to-sink model explain why it is sweet?' This connects to real Kenyan agricultural experience and previates further organic chemistry content.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed a ring-barked (girdled) bean or sweet-potato stem where a band of outer tissue (phloem) was removed: tissue swelled and accumulated just ABOVE the cut but not below. We also analysed Evidence Cards showing sieve tubes, companion cells, and aphid stylet experiments confirming sugary sap flows under pressure through phloem.
What did I learn?	Plants have TWO transport systems. Xylem carries water and mineral salts upward from roots to leaves (as shown by the red-dye experiment in Lessons 1–3). Phloem carries sugars and other assimilates by TRANSLOCATION from SOURCE regions (mature photosynthesising leaves) to SINK regions (roots, developing maize cob, young growing leaves) — the direction is always from where sugar is made or stored toward where it is needed. Ring-barking blocks phloem but not xylem, explaining why girdled trees eventually die even though water still reaches their leaves.
How does this explain the phenomenon?	The red-stained rings in our Kisumu maize stem cross-section (the anchoring phenomenon) are XYLEM — they carry only water and mineral salts upward. The colourless outer tissue of the vascular bundle is PHLOEM — it carries sugars made by photosynthesis to the rest of the plant by translocation. Together, these two systems explain how the maize plant sustains every

	living cell: xylem supplies water and minerals for photosynthesis; phloem distributes the glucose produced. The polythene-bag water droplets (transpiration, second part of the phenomenon) will be fully explained in Lesson 5 — that water exits through stomata from the xylem stream.
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LESSON 5 (80 minutes): Transpiration Experiment Setup: Designing Investigations into Factors Affecting Water Loss in Plants

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners design and set up controlled experiments to investigate how specific factors — leaf surface area (petroleum jelly), wind, humidity, and light intensity — affect the rate of transpiration in plants, thereby advancing their understanding of how water that travels upward through xylem ultimately exits the maize plant.
Knowledge	Learners will know that transpiration is the loss of water vapour from plant surfaces (mainly stomata); that environmental factors (wind, light, humidity) and structural factors (leaf surface area with functional stomata) influence its rate; and that a controlled investigation requires a hypothesis, identified independent/dependent/controlled variables, and an initial baseline measurement.
Skills	Learners will be able to: (a) state a testable hypothesis for their assigned factor; (b) correctly identify and operationalise independent, dependent, and controlled variables; (c) set up a transpiration experiment safely using locally available materials; (d) record initial measurements systematically in a results table; (e) apply SEP: Planning and Carrying Out Investigations.
Attitudes	Learners will appreciate the value of fair testing and collaborative roles in scientific inquiry; demonstrate unity by sharing equipment responsibly; and show responsibility by observing all safety precautions during setup.
Key Inquiry Question	Why is transport in plants important?
Purpose in Storyline	In Lessons 1–4 learners saw red dye travel upward through xylem and water droplets form inside a sealed polythene bag. They now know water moves up through xylem and exits via transpiration. But the driving question asks: what happens to that water in the end — and what controls how fast it leaves? Lesson 5 shifts from observing a teacher-led demonstration to learners themselves designing investigations. Each group manipulates one factor and sets up their experiment, making this the evidence-gathering engine that Lessons 6 and 7 will analyse and explain.
Safety Notes	Petroleum jelly is non-toxic but keep away from eyes — wash hands after application. Fans (if electric) must be kept away from water sources; confirm cables are intact before use. Polythene bags must not be placed over learners' heads at any time. Scissors used to trim leaves must be handled point-down and stored blade-inward. No learner should stand on furniture to adjust light sources. Teacher confirms all setups before learners leave the room.

B. LESSON OVERVIEW

Groups of 4–5 learners each investigate one factor affecting transpiration using freshly cut leafy stems (succulent cuttings such as sweet potato or bean, mirroring the polythene-bag phenomenon). Group A covers leaf undersurfaces with petroleum jelly (reducing functional stomata). Group B directs a fan at their shoot (increasing wind speed). Group C encloses their shoot in a sealed clear polythene bag (raising humidity). Group D varies light intensity by moving the setup from full Kisumu afternoon sunlight to shade. All groups state a hypothesis, identify variables, measure initial stem mass or initial water volume in the beaker, and sketch their setup. Safety rules are reviewed before any equipment is touched. The lesson closes with each group posting their hypothesis and variable table on the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Divisibility Rules to Find Factors (Flexbook) Source: CK-12  Search ARES for similar readings	Learners individually write a prediction in their science journals answering: 'Your group will change ONE thing about the environment or the leaf surface of a freshly cut sweet-potato shoot. Before you touch any equipment, predict — will your change make the plant lose water FASTER, SLOWER, or the SAME? Write one sentence explaining WHY you think so, using what you know about stomata and the polythene-bag observation from Lesson 3.' After 4 minutes of individual silent writing, pairs share their reasoning for 2 minutes. The teacher then cold-calls six learners (two per factor group) to voice predictions aloud, and the class votes silently (thumbs up/down/sideways) on whether they agree.	1. Display the anchoring phenomenon image (polythene bag with condensation droplets and the red-stained xylem cross-section) on the board. Say: 'Look at this bag. Water got out of the plant even though no liquid water was added. Today YOU are going to find out what makes that process faster or slower — but first, predict.' 2. Issue the prediction prompt on a half-sheet or project it. Allow 4 minutes of SILENT individual writing. Do NOT accept shouted answers during this time. 3. After 4 minutes say: 'Turn to your partner. You have 90 seconds. Tell them your prediction AND one piece of evidence from our earlier lessons that supports it.' 4. After partner talk, cold-call SIX learners: 'Amina — what did you predict for Group B, the fan group, and why?' (WAIT 12 seconds for response.) Repeat for five more learners from different groups. 5. Conduct a silent vote: 'Thumbs up if you agree with Amina's reasoning, sideways if unsure, down if you disagree.' Note distribution on board without affirming or correcting — say: 'We will test these ideas today. Hold your prediction — we will revisit it at the end.'	Individual Predict-Then-Share (a variation of Think-Pair-Share anchored to the phenomenon). This activates prior knowledge about stomata and transpiration from Lessons 3–4, surfaces existing misconceptions (e.g., 'petroleum jelly blocks ALL water loss'), and creates cognitive dissonance that motivates careful experimental design. Recording the prediction in writing externalises thinking so learners can compare it to results in Lesson 6.	Teacher circulates during silent writing and reads over shoulders. Observable indicators of understanding: learner mentions stomata, water vapour, or evaporation in their prediction rationale. Red flag: learner predicts with no mechanistic reason (e.g., 'it will be faster because of the fan' with no link to evaporation rate or stomata). Teacher notes 2–3 such learners by name for targeted questioning during Phase 3.
Observe Phase  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Groups move to their pre-labelled bench stations. Each station has: a freshly cut leafy stem (sweet potato or bean cutting, approximately 15 cm, cut under water to prevent air embolism), a 250 ml beaker of water with a thin layer of cooking oil on top (to reduce evaporation from the water surface — a controlled variable), a	1. Before any group touches equipment, call the class to attention: 'FREEZE. Before we set up, let us agree on THREE non-negotiables for today.' Read aloud: (a) Petroleum jelly stays on leaves only — not on skin or eyes; wash hands immediately after application. (b) The fan cable must NOT be near the water beakers.	Structured Experimental Design Protocol with Phenomenon Anchoring. Before recording any data, each group must complete the variable table and write a hypothesis in the IF-THEN-BECAUSE format. This structure (adapted from NGSS SEP: Planning and Carrying Out Investigations) prevents learners	Teacher collects the variable tables from each group at the end of this phase (or photographs them with a phone). Observable indicators: (1) Independent variable is correctly identified as the ONE factor the group is changing. (2) Dependent variable is stated as a measurable quantity (water volume lost / mass change),

 HTML: Divisibility Rules to Find Factors (Flexbook) Source: CK-12  Search ARES for similar readings	balance or measuring cylinder for initial mass/volume, and group-specific equipment. – Group A (Petroleum Jelly): Learners use a cotton bud to coat the UNDER surface of every leaf on their stem with petroleum jelly, then place the stem in their beaker. They record the number of leaves coated and the initial water volume. – Group B (Wind / Fan): Learners set up their stem in the beaker at a fixed distance (30 cm) from a small table fan. Fan is switched on at the lowest setting. They record initial water volume and air temperature at the fan outlet using a thermometer. – Group C (Humidity / Polythene Bag): Learners place a clear polythene bag loosely over the shoot (NOT the beaker) and seal it at the stem with a rubber band, leaving the stem in water. They record initial water volume and note starting bag clarity. – Group D (Light Intensity): Learners position one setup in direct afternoon sunlight near the window and one in the shade of the corridor, recording initial water volume and light intensity category (full sun / shade). All groups fill in a pre-printed variable table (Independent Variable / Dependent Variable / Controlled Variables) and write their hypothesis in the format: 'If [independent variable changes], then [dependent variable] will [increase/decrease/stay the same]	(c) Polythene bags are NEVER placed over anyone's face or head. Ask: 'Who can tell me why these rules connect to our experiment's validity as well as our safety?' Cold-call TWO learners. (WAIT 10 seconds each.) 2. Distribute equipment to bench captains only. Say: 'Bench captain, do NOT distribute until your group has agreed on who does each role: one measurer, one recorder, one apparatus handler, one safety monitor.' Allow 2 minutes for role assignment. 3. Circulate every 4–5 minutes. At each group, ask ONE probing question: — Group A: 'Why are you coating the UNDER surface and not the top? What do you remember about where stomata are most concentrated?' (WAIT 12 seconds.) — Group B: 'What are you keeping the same between your setup and Group A's setup so we can compare results next lesson?' (WAIT 12 seconds.) — Group C: 'What do you think will happen to the air inside the bag as time goes on? How might that slow transpiration?' (WAIT 15 seconds.) — Group D: 'How will you make sure the stems are otherwise identical — same number of leaves, same stem length?' (WAIT 12 seconds.) 4. At the 20-minute mark, call a 2-minute whole-class pause: 'Pens down. Each group safety monitor — stand up and check: is any cable near water? Is petroleum jelly off hands? Thumbs up when clear.' Wait for all thumbs before	from 'just following steps' and forces them to articulate the mechanistic link between their manipulated factor and the transpiration rate — directly connecting to the driving question about what happens to the water that travels up through xylem.	not a vague term like 'transpiration amount'. (3) At least three controlled variables are listed (e.g., same species, same stem length, same number of leaves, same time period, same starting water volume). Red flag: group lists the factor they are changing as BOTH independent AND controlled variable, or lists 'transpiration' as both IV and DV. Teacher addresses this immediately before the group leaves setup.
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	because [mechanistic reason].' Groups sketch their apparatus setup with labels.	continuing. 5. At the 28-minute mark: 'Record your INITIAL measurement NOW. You have 90 seconds. This is your Time = 0 data point. Without it, you have no experiment.'		
Explain Phase  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Divisibility Rules to Find Factors (Flexbook) Source: CK-12  Search ARES for similar readings	Groups rotate to a Gallery Walk of the four bench setups (2 minutes per station). At each station a card shows the group's hypothesis and variable table (posted by the bench captain). Learners use sticky notes to leave one 'I notice...' comment and one 'I wonder...' question on each card. After the gallery walk, the class reconvenes. The teacher selects four 'I wonder' questions (one per group) that best connect to the driving question and asks the originating group to respond. Learners then update their own group's variable table if they received a useful challenge on their sticky notes. Finally, each learner individually answers in their journal: 'Explain, in two sentences, how your group's experiment will give us evidence about what happens to the water that travels up through xylem in the maize plant on our Kisumu farm.'	1. Instruct gallery walk: 'You have 8 minutes total — 2 minutes per station. Read the hypothesis and variable table. Leave ONE sticky note starting with 'I notice...' and ONE starting with 'I wonder...'. Be specific — write the group letter on your note.' Use a timer visible to all. 2. After the gallery walk, say: 'I am going to read four 'I wonder' questions — one about each group's setup. The group whose setup it is will respond. You have 30 seconds to confer with your group before answering.' Read the selected questions one at a time. After each group confers, cold-call the RECORDER (not the bench captain) to answer. (WAIT 12 seconds after question is read before starting group confer time.) 3. After responses, prompt: 'Did any sticky note on YOUR station make you want to change your variable table? You have 90 seconds to update it if needed.' 4. Direct individual journal writing: 'On your own — two sentences only — connect YOUR experiment to the maize plant and our driving question. Start with: The water that travels up through xylem in a Kisumu maize plant eventually...' Allow 3 minutes. 5. Cold-call THREE learners to read their sentences. After each, ask the class: 'Does this sentence use evidence from our lessons, or is it just a claim?' (WAIT 10	Gallery Walk with Sticky-Note Critique followed by Phenomenon Re-anchoring Prompt. The gallery walk uses peer-generated questions to stress-test each group's experimental design before any data is collected — a lower-stakes form of peer review that mirrors scientific practice. The individual journal sentence forces each learner to personally connect their small-group experiment back to the whole-plant, whole-farm level of the driving question, preventing the common error of treating transpiration as an isolated lab activity rather than as the endpoint of the water-transport pathway they have been tracing since Lesson 1.	Teacher reads the individual journal sentences while circulating during writing (3 minutes is sufficient to scan 30 sentences). Observable indicator of meeting expectations: sentence names transpiration as water leaving via stomata AND links it to the xylem-transport pathway studied in Lessons 2–4. Red flag: sentence describes transpiration as 'water dripping off leaves' or does not mention stomata or vapour — teacher flags these learners for a brief clarifying conversation before Lesson 6.

		seconds for hands before selecting a responder.)		
Driving Question Board (DQB) Creation  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Divisibility Rules to Find Factors (Flexbook) Source: CK-12  Search ARES for similar readings	Each group receives two DQB cards: one ORANGE card on which they write their hypothesis ('We think [factor] will [increase/decrease] transpiration because...') and one BLUE card on which they write their biggest remaining question about their experiment or about transpiration in general. Groups post both cards on the class DQB under the column header 'What we are testing — Lesson 5.' The class then does a 3-minute silent read of all posted cards. Learners place a small dot sticker on the ONE blue question they most want answered by Lesson 6. The teacher records the top-voted question on a separate 'priority question' strip at the top of the DQB.	1. Distribute orange and blue cards. Say: 'Orange card: your group's hypothesis — write it in the IF-THEN-BECAUSE format, exactly as it appears on your variable table. Blue card: the one thing you most need to know to interpret your results next lesson. You have 3 minutes.' 2. Groups post cards simultaneously. Say: 'Bench captains, post your cards now. Stick them under Lesson 5 — hypotheses on the left, questions on the right.' 3. Direct silent read: 'Stand up, walk to the DQB, read ALL eight cards in silence. You have 3 minutes. No talking.' 4. Distribute dot stickers (one per learner): 'Put your dot on the ONE blue question that YOU most want us to answer in Lesson 6. One dot only.' 5. Tally dots. Read the winning question aloud. Say: 'This becomes our priority question for Lesson 6. Write it at the top of your journal page for next lesson.' 6. Gesture to the full DQB and say: 'Look at how our board has grown. In Lesson 1 we asked why red dye only went through certain channels. Today we are testing four different ways that water leaves the plant at the other end of that same journey. Our driving question is getting closer to a full answer — but we need data first.'	Hypothesis Posting and Democratic Prioritisation on the DQB. Making hypotheses public creates accountability — groups know their prediction will be compared to real data in Lesson 6. The dot-voting ritual gives learners agency over the inquiry direction and models how scientific communities prioritise research questions. Connecting the DQB growth to the original phenomenon (red-dye channels → transpiration endpoint) keeps the storyline coherent across the 8-lesson sequence.	Teacher photographs the full DQB after posting. Observable indicators: orange cards use IF-THEN-BECAUSE format; the named independent variable matches the group's assigned factor; blue question cards are mechanistic (e.g., 'Will the bag eventually become so humid that transpiration stops completely?') rather than procedural (e.g., 'How long do we wait?'). Red flag: a group's orange card states a hypothesis that is not falsifiable by their experimental setup — teacher returns the card with a prompt: 'How will you measure this with the equipment at your station?'
Model Building Phase  VIDEO:	Learners individually update the cumulative plant-transport model they have been building since Lesson 1. Today's addition: on	1. Say: 'Open your cumulative model — the diagram we have been adding to since Lesson 1. Today you add the transpiration	Cumulative Model Revision (individual, annotated diagram). The model has served as a running record of the class's	Teacher collects models for a 5-minute review between lessons (or photographs them). Observable indicators: stomata are drawn on

Photorespiration Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Divisibility Rules to Find Factors (Flexbook) Source: CK-12 Search ARES for similar readings	their model diagram of a maize plant, each learner draws and labels the STOMATA on the leaf surface and adds arrows showing water vapour leaving through the stomata. They add a second annotation layer showing their group's factor — e.g., Group A draws petroleum jelly blocking stomata pores; Group B draws wind arrows accelerating vapour removal; Group C draws humidity arrows slowing vapour gradient; Group D draws a sun/shade symbol affecting stomata opening. Learners write one sentence under their model: 'My group's experiment will change the rate of transpiration by changing _____, which affects _____ in the stomata/environment.'	endpoint. First, everyone draws stomata on the leaf surface — small pores on the underside. Add arrows showing water vapour leaving. You have 2 minutes.' (WAIT and circulate. Remind learners: 'Stomata are mostly on the LOWER epidermis — check your notes from Lesson 3.') 2. After 2 minutes: 'Now add YOUR group's factor to the model. Show HOW it changes transpiration using arrows, symbols, or a small diagram. You have 3 minutes.' 3. Cold-call ONE learner from each of the four groups to hold up and briefly explain their model addition (30 seconds each): 'Otieno — show us where you drew the petroleum jelly and explain what it does to the water vapour.' (WAIT 10 seconds.) Repeat for Groups B, C, D. 4. Say: 'Write your one-sentence annotation now. It must name your independent variable AND name the mechanism — stomata, evaporation gradient, or stomata aperture.' Allow 2 minutes. 5. Close: 'In Lesson 6, we collect our data. In Lesson 7, we use data from ALL four groups to build a complete explanation. Keep your model safe — you will need it.' Point to the driving question on the wall: 'We started by asking where water goes after it reaches the topmost leaves of a Kisumu maize plant. Today you designed experiments to find out. Next lesson — the evidence.'	growing understanding since Lesson 1 (xylem pathway → osmosis → translocation → transpiration). Adding the factor-specific annotation today ensures each learner personally represents the mechanistic link between their experiment and the broader transport pathway — preventing compartmentalised thinking where 'the experiment' and 'the model' are treated as separate activities. Sharing model additions via cold-call creates a rapid whole-class synthesis of all four experimental factors.	the LOWER leaf surface (not upper); water vapour arrows exit through stomata (not through the leaf blade generally); the group-specific factor is mechanistically represented (e.g., Group A shows blocked pores, not just a label 'petroleum jelly'). Red flag: learner draws water dripping from leaf edges rather than vapour through stomata — misconception that transpiration is liquid dripping, not vapour diffusion. Teacher prepares a targeted correction prompt for the start of Lesson 6.
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) **HYPOTHESIS QUALITY** — Did all four groups write hypotheses in the IF-THEN-BECAUSE format with a mechanistic reason, or did groups default to vague predictions? If mechanistic reasoning was weak, plan to open Lesson 6 with a whole-class review of one strong and one weak hypothesis (anonymised) before data collection begins. (2) **VARIABLE IDENTIFICATION** — Were learners able to distinguish between independent, dependent, and controlled variables, or did any group list their factor as both IV and a controlled variable? This is a high-frequency misconception; if it appeared in more than one group, insert a 5-minute variable-sorting activity at the start of Lesson 6 using a worked example (e.g., 'In an experiment on ugali cooking time, what is the IV if we change flame size?'). (3) **SAFETY COMPLIANCE** — Were all three non-negotiable safety rules observed throughout? Note any specific incidents for the school safety log and adjust bench layout if water and electrical equipment were too close. (4) **DQB ENGAGEMENT** — Did the dot-voting produce a genuine priority question, or did learners place dots randomly? If the winning question is procedural rather than conceptual, reframe it as a conceptual question for the start of Lesson 6 and explain to learners why conceptual questions drive science forward. (5) **PHENOMENON CONNECTION** — When learners wrote their two-sentence journal entries, did they connect their experiment back to the maize plant and the driving question, or did they describe their experiment in isolation? If fewer than 70% of sentences made the connection, dedicate the first 5 minutes of Lesson 6 to re-reading the driving question together and explicitly mapping each group's factor onto the whole-plant diagram.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A freshly cut sweet-potato or bean stem was placed in a beaker of water. Different groups changed one factor each: Group A coated leaf undersurfaces with petroleum jelly; Group B directed a fan at the shoot; Group C enclosed the shoot in a sealed clear polythene bag; Group D placed their setup in full Kisumu afternoon sunlight versus shade. Each group recorded the initial water volume at Time = 0.
What did I learn?	Transpiration — the loss of water vapour from plant surfaces — can be affected by leaf surface conditions (open or blocked stomata), wind speed, surrounding humidity, and light intensity. A fair test requires changing only ONE variable at a time while keeping all others constant. The rate of transpiration is measured by how much water the plant loses over time.
How does this explain the phenomenon?	Stomata are small pores, mostly on the underside of leaves, through which water vapour diffuses out of the plant into the surrounding air. The rate of this diffusion depends on the concentration gradient of water vapour between the inside of the leaf and the air outside. Anything that increases that gradient (wind removing vapour, bright light opening stomata wider) speeds transpiration; anything that reduces it (high humidity in a sealed bag, petroleum jelly blocking pores) slows it. This is the final stage of the water journey that began when roots absorbed water from the Kisumu farm soil and xylem carried it to the topmost leaves.

LESSON 6 (40 minutes): Transpiration Data Collection and Analysis: What Patterns Does Our Evidence Reveal?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners collect final transpiration measurements from their group experiments, pool class data, analyse patterns in graphs, and use evidence to explain how environmental factors control the rate of water loss from plant surfaces.
Knowledge	Learners will be able to: (1) state how wind speed, humidity, light intensity, and blocked stomata each affect the rate of transpiration; (2) explain the relationship between the stomatal gradient and the direction of water vapour movement; (3) link experimental findings back to the polythene-bag observation from Lesson 1.
Skills	Learners will be able to: (1) record transpiration measurements accurately in a data table; (2) construct and interpret bar graphs and line graphs from class-pooled data; (3) identify patterns and anomalies in data sets; (4) use evidence to support or refute their earlier hypotheses.
Attitudes	Learners demonstrate scientific integrity by recording actual results even when they differ from their hypothesis; they show collaborative respect when pooling and comparing group data.
Key Inquiry Question	Why is transport in plants important? — This lesson advances the answer by showing that transpiration is not random: its rate is controlled by measurable environmental conditions, which in turn affects how efficiently a maize plant on a Kenyan farm can pull water from the soil to its topmost leaves.
Purpose in Storyline	In Lesson 5 learners designed and set up transpiration experiments. Lesson 6 is the evidence-collection and analysis lesson: groups read their final measurements, pool data on the class board, draw graphs, and identify patterns. This evidence will feed directly into Lesson 7, where the significance of transpiration for plant survival and Kenyan water cycling will be explained.
Safety Notes	Handle plants and measuring equipment gently. Wash hands after handling petroleum jelly or any plant material. Ensure fans used in the wind experiment are placed on stable surfaces and kept away from water. No direct eye contact with bright lamps used in the light-intensity experiment.

B. LESSON OVERVIEW

Groups collect their final water-loss measurements from the Lesson 5 set-ups, enter data into a class table on the board, and draw graphs. Through guided sensemaking the class identifies which factor caused the greatest increase or decrease in transpiration rate and explains the mechanism behind each pattern. The lesson closes by reconnecting to both the polythene-bag phenomenon and the driving question, and learners update their evidence on the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photorespiration Source: Khan	Before reading their final measurements, each learner individually writes two sentences in their science notebook: (1) a	Teacher opens with a brief recap hook: 'Cast your mind back to Lesson 1 — we saw water droplets collecting inside a polythene bag	Prediction-before-data routine creates cognitive commitment so that learners experience the productive tension of comparing	Teacher scans notebooks during circulation to check whether learners are using mechanistic reasoning (e.g., referencing

Academy (English - US curriculum) Search ARES for similar videos  HTML: Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 Search ARES for similar readings	ranked prediction of which experimental group they expect showed the HIGHEST transpiration rate and why, and (2) a ranked prediction of which group showed the LOWEST rate. Learners refer to the hypothesis they wrote in Lesson 5 and circle whether they still agree with it or have changed their mind based on what they have observed qualitatively over the experiment period. Pairs briefly share their rankings with each other (1 minute).	sealed around a leafy branch in the afternoon sun here in Kisumu. Today we finally have the numbers to explain EXACTLY why that happened and what controls how fast it happens.' Teacher then says: 'Before we read a single number, I want you to make one last prediction. In your notebook, rank the four experimental conditions — blocked stomata, high wind, high humidity, and bright light — from the one you expect lost the MOST water to the one that lost the LEAST. Write your reason in one sentence.' Allow 90 seconds of silent individual thinking. Teacher circulates and notes two or three contrasting predictions to revisit after data collection.	prediction to evidence — a core scientific practice. Recalling the polythene-bag phenomenon re-anchors the lesson in the real-world context.	stomata, concentration gradient, or water vapour) rather than purely intuitive ranking. This informs the depth of prompting needed during the Explain phase.
Observe Phase  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 Search ARES for similar readings	Step 1 — Final Measurements (8 minutes): Each group carefully reads and records the final mass (or water level, depending on their set-up) of their potometer or sealed-bag apparatus. They subtract the initial measurement recorded in Lesson 5 to calculate water lost in grams or millilitres. They also note qualitative observations: Did leaves wilt? Did condensation form? Was there visible change in soil moisture? They enter all values into their group data table using the template provided in Lesson 5. Step 2 — Class Data Pool (4 minutes): One representative from each group writes their group result on the class data table drawn on the board. The table has five columns: Experimental condition Initial mass (g) Final mass (g) Water lost (g) Rate (g per 10 min). Teacher or a nominated learner completes the	During Step 1, teacher moves from group to group asking: 'Has the result surprised you? How does it compare to your Lesson 5 hypothesis?' [WAIT TIME: after each question, pause for 10 seconds before prompting further.] During Step 2, teacher reads aloud each group result as it goes on the board and asks the class: 'Does anyone want to question this number before we accept it into our class table? Could it be an outlier?' This models scientific peer review in a low-stakes way. During Step 3, teacher demonstrates on the board how to choose the correct graph type: 'We use a bar graph when our independent variable is CATEGORICAL — like type of condition. We use a line graph when it is CONTINUOUS — like light intensity measured in lux. Which applies to your group?' [WAIT TIME: 15 seconds for learners to decide before teacher	Pooling data across groups creates a richer evidence base than any single group could generate — mirroring how scientists share findings. Drawing graphs by hand reinforces the link between numbers and visual patterns.	Teacher checks that axes are correctly labelled with units and that the dependent variable (water lost in g or rate in g per 10 min) is always on the y-axis. Common error to watch for: learners plotting initial mass rather than change in mass.

	rate column together as a class by dividing water lost by the number of 10-minute intervals the experiment ran. Step 3 — Graph Construction (6 minutes): Each learner individually draws either a bar graph (comparing the four conditions) or a line graph (if the light-intensity group has multiple data points at different lux levels) in their notebook. Axes must be labelled with units. Learners use the class-pooled data so that every individual has the full picture.	confirms.]		
Explain Phase  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12  Search ARES for similar readings	Step 1 — Pattern Identification (4 minutes): Using their completed graphs, learners answer three analysis questions in their notebooks: (Q1) Which condition produced the highest transpiration rate? (Q2) Which condition produced the lowest rate? (Q3) Describe the trend — as wind speed increases / as humidity increases / as light increases, what happens to the rate? Each group is assigned the question most relevant to their experiment. Step 2 — Mechanism Discussion (5 minutes): Groups discuss the MECHANISM behind their pattern using a sentence stem on the board: 'The rate of transpiration was [higher/lower] under [condition] because [stomata / concentration gradient / water vapour / diffusion / guard cells]...' A representative from each group shares their sentence with the class. Teacher facilitates by linking all explanations back to the water vapour concentration gradient between leaf air spaces and the surrounding air — the single unifying mechanism. Step 3 — Reconnecting to the Phenomenon (3 minutes): Teacher directs	For Q3 pattern analysis, teacher prompts: 'Do not just say the rate went up — tell me WHY by thinking about what is happening at the stomata.' [WAIT TIME: 20 seconds.] During mechanism sharing, if a group says wind increases transpiration, teacher asks the class: 'What exactly does wind DO at the leaf surface that changes the gradient? Turn and tell your partner.' [WAIT TIME: 15 seconds for paired discussion before cold-calling.] For the polythene-bag reconnection, teacher says: 'The bag was sealed and no liquid water was added. Where did the water vapour come from? Your data from the sealed-bag group might help you answer that.' Teacher avoids giving the answer immediately — instead writes on the board: 'Source of water vapour = ?' and lets learners fill it in. Key vocabulary to reinforce orally and in writing: transpiration pull, concentration gradient, stomata, water vapour, diffusion, guard cells. Teacher uses local context: 'Think about a maize field in Siaya County on a hot, windy afternoon after a dry spell — based on our data, what is	The sentence-stem strategy scaffolds mechanistic explanation without removing cognitive challenge. Reconnecting to the Lesson 1 polythene-bag phenomenon creates a satisfying closure loop — learners see that they now have evidence-based answers to a question that was puzzling in Lesson 1. Local Kenyan agricultural context makes the science personally meaningful.	Teacher collects the two-sentence polythene-bag explanation as an exit-quality check. Look for: (a) learner identifies the leaf as the source of water vapour, (b) learner uses the word stomata or diffusion, (c) learner links the bag sealing to the rise in humidity reducing the gradient. Learners who cannot meet criteria (a) or (b) will need targeted support in Lesson 7.

	attention to a photograph or sketch of the polythene-bag observation from Lesson 1 posted on the wall. Teacher asks: 'The bag was sealed — so why did water droplets still form inside? Use your data as evidence.' Learners write a two-sentence explanation in their notebooks.	happening to the water that farmer Atieno irrigated that morning?'		
Driving Question Board (DQB) Creation  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Formulas for Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12  Search ARES for similar readings	Each group receives two sticky notes of different colours. On the FIRST note (green), they write one question from the DQB that their experimental data has now answered — and they write the evidence-based answer in one sentence. On the SECOND note (orange), they write one NEW question their data raised that is still unanswered. Groups take turns posting their notes on the DQB. The teacher reads aloud the answered questions and asks the class to vote by show of hands whether they agree the question is now sufficiently answered. Unanswered questions are flagged for Lesson 7.	Teacher says: 'Scientists do not just collect data and put it away — they ask what their data means for the next question. Look at our DQB from Lesson 1. Point to one question that our graphs today can now answer.' After groups post, teacher facilitates a brief gallery walk (1 minute, standing at desks and looking at the board) then says: 'I can see several groups have asked WHY transpiration matters if it means the plant is losing water — that is EXACTLY what Lesson 7 will tackle. Hold that question.' Teacher uses a contrasting-cases move: 'Group A says high wind INCREASES water loss. Group C says high humidity DECREASES it. How can BOTH be true at the same time — what is the variable they are both affecting?' [WAIT TIME: 20 seconds before accepting responses.]	The two-colour sticky note protocol makes the distinction between answered and open questions visible and concrete. The contrasting-cases prompt (wind vs. humidity) deepens understanding by showing that both variables act on the same underlying mechanism — the concentration gradient.	Teacher scans the answered-question notes for quality of evidence citation. A strong note will reference a specific measurement (e.g., 'the sealed-bag group lost only 0.3 g versus 1.8 g in the open group, so humidity reduces transpiration by narrowing the concentration gradient'). Vague notes are verbally prompted: 'Can you add a number from our class table to support that?'
Model Building Phase  VIDEO: Photorespiration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Formulas for	In the final 4 minutes, each learner adds a NEW layer to the running annotated diagram they began building in earlier lessons. Using a coloured pen different from previous lessons, they annotate the leaf region of their diagram to show: (1) an open stoma with an arrow indicating the direction of water vapour movement; (2) a label showing the high-concentration water vapour region	Teacher circulates and prompts: 'Show me on your diagram where the concentration gradient exists — draw a dotted line between the two regions.' [WAIT TIME: 10 seconds.] For extension learners, teacher asks: 'If transpiration rate increases on a hot Kisumu afternoon, what does that mean for the rate at which water and dissolved mineral salts are moving up the xylem of a maize stem?'	Incremental model building across lessons creates a cumulative, visible knowledge artifact. Using a different colour for each lesson update makes the progression of understanding tangible and helps learners see how each lesson adds a new explanatory layer.	Teacher checks that the concentration gradient is correctly drawn — high inside the leaf, lower outside. Common misconception to catch: learners who draw the arrow pointing INTO the leaf, suggesting water vapour enters rather than leaves. This is corrected in real time by asking: 'Which side has more water vapour — the air space inside the leaf or the dry air outside on a

Problem Solving: Finding Distance, Rate, and Time (PLIX) Source: CK-12 Search ARES for similar readings	inside the leaf air space and the lower-concentration region outside; (3) a note showing how wind, humidity, and light each affect this gradient, with a plus or minus symbol to indicate whether each factor increases or decreases the rate. Learners who finish early add a note connecting transpiration rate back to the speed at which water is pulled up the xylem from the roots — previewing the cohesion-tension content of Lesson 7.	Add an arrow to show that link.' Teacher closes the lesson with a forward-bridge statement: 'Today you have proved with real measurements that environmental conditions control how fast water leaves a plant. In Lesson 7 we will ask — is transpiration a problem for the plant, or is it actually essential? And we will explore what farmers around the Mau Forest need to know about this.' Teacher thanks learners for scientific integrity in recording honest results even when they differed from predictions.	sunny Kisumu afternoon?'
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did all four experimental groups obtain plausible data, or did any set-up fail to show a measurable difference? If so, why — and how will you address this in Lesson 7 discussion? (2) Were learners able to articulate the concentration gradient mechanism in their own words, or did they rely only on memorised phrases? (3) Did the polythene-bag reconnection feel natural, or did learners still seem puzzled? Plan a brief revision of diffusion and water vapour for the opening of Lesson 7 if needed. (4) Note which learners posted strong evidence-cited sticky notes on the DQB — these learners can serve as peer explainers in Lesson 7 group work. (5) Consider whether the graph-drawing time was sufficient — if many learners did not finish, allow 5 minutes at the start of Lesson 7 for completion before moving to significance discussion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the group with high wind lost the most water, the group with blocked stomata (petroleum jelly) lost the least, the group in the sealed polythene bag lost very little, and the group under bright light lost more than the group in shade. We pooled all four group results into one class data table and drew bar graphs showing these differences.
What did I learn?	We learned that the rate of transpiration is controlled by the concentration gradient of water vapour between the inside of a leaf and the surrounding air. Wind increases the rate by sweeping away water vapour outside the stomata, restoring a steep gradient. High humidity reduces the rate by raising the water vapour concentration outside, flattening the gradient. Bright light increases the rate by causing stomata to open wider. Blocking stomata reduces the rate almost to zero.
How does this explain the phenomenon?	We explained why water droplets formed inside the sealed polythene bag in Lesson 1: the leaf was continuously losing water vapour through its stomata by diffusion down the concentration gradient, and when the bag trapped that vapour it condensed on the cool inner surface. This shows that transpiration is a continuous, measurable process driven by physical gradients — not a random event — and its rate changes predictably when environmental conditions change.

LESSON 7 (80 minutes): Importance of Transpiration and Community Application

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To explain the significance of transpiration in plant life and communicate its importance to the community through a farmer-focused awareness message.
Knowledge	Learners will be able to: (i) describe the significance of transpiration — cooling of leaves, generation of transpiration pull (cohesion-tension theory), and distribution of mineral salts; (ii) explain the role of transpiration in local and regional water cycling, with reference to the Mau Forest catchment; (iii) explain the importance of proper watering, manuring, and responsible fertiliser use (DAP, CAN, Urea) for sustaining crop growth.
Skills	Constructing explanations that link cohesion-tension theory to the upward movement of water observed in the red-dye phenomenon; obtaining, evaluating, and communicating information through drafting a community awareness message for farmers.
Attitudes	Learners appreciate the role of transport in plants and show patriotism by sensitising the community on the importance of watering and manuring plants to sustain plant growth, and by creating awareness on the responsible use of fertilisers (Di-ammonium Phosphate, Calcium Ammonium Nitrate, Urea) in boosting crop production.
Key Inquiry Question	Why is transport in plants important?
Purpose in Storyline	In Lessons 1–6 learners gathered evidence — the red-dye maize stem showed water travelling upward through xylem, while the polythene-bag experiment revealed water vapour escaping from leaves. Lesson 7 now asks: WHY does the plant lose water through transpiration at all? Learners synthesise all prior evidence to explain the three key functions of transpiration (cooling, transpiration pull, mineral distribution), connect those functions to the Mau Forest water cycle, and translate their understanding into a practical community message for farmers in the Kisumu region — directly advancing the driving question by accounting for what ultimately happens to the water after it reaches the topmost leaves.
Safety Notes	No hazardous chemicals are used in this lesson. If printed fertiliser packets (DAP, CAN, Urea) are brought in as realia, learners must not open the packets; hands should be washed after handling. Ensure learners are seated safely during whole-class and group discussion phases. Supervise use of any digital devices according to school ICT policy.

B. LESSON OVERVIEW

Lesson 7 is the EXPLAIN lesson in the 8-lesson sequence on Transport in Plants. Building on the red-dye maize stem evidence (xylem channels) and the polythene-bag transpiration evidence from earlier lessons, learners now synthesise WHY transpiration occurs and why it matters — for the individual maize plant, for Kenyan farms, and for the wider environment including the Mau Forest catchment. The lesson opens with a brief predict-and-recall warm-up, moves through a teacher-facilitated whole-class discussion to construct explanations for the three functions of transpiration, then shifts to small-group work where learners research and discuss the environmental importance of transpiration in local water cycling. The lesson culminates in a community action task: each group drafts a short Swahili/English awareness message for farmers on proper watering, manuring, and responsible use of DAP, CAN, and Urea fertilisers. The Driving Question Board is revisited and updated. SEPs foregrounded: Constructing Explanations; Obtaining, Evaluating, and Communicating Information.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Hydrogen bonding in water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES for similar readings	Learners individually write a 2-minute 'exit prediction' in their notebooks: 'The maize plant keeps losing water through its leaves even on a hot afternoon in Kisumu. Give TWO reasons why you think this water loss might actually be USEFUL to the plant — not just wasteful.' They then share predictions with a shoulder partner before a whole-class share-out.	1. Display the anchoring phenomenon images side-by-side on the board or screen: (A) the cross-section of the red-dye maize stem showing vivid red rings in the xylem, and (B) the polythene bag with water droplets on its inner surface. Say: 'Look at these two images we have been investigating all week. Image A shows us HOW water travels up — through the xylem. Image B shows us that water eventually LEAVES the plant through the leaves. Today's big question is: WHY does the plant allow this water to escape? Is it a mistake — or is it useful?' (WAIT 10 seconds for silent individual thinking.) 2. Give the written prediction task. Allow 2 minutes of silent individual writing. 3. Cold-call 4 learners (select: 1 high-confidence, 1 mid-confidence, 2 learners who rarely volunteer — use the class register to track). Record their predictions verbatim on the board in two columns: 'Water loss is useful because...' 'Water loss might be harmful because...'. Do NOT evaluate — say: 'We are not judging yet — we are collecting ideas to test against evidence today.' 4. Leave predictions visible for the entire lesson so learners can revisit them.	Elicit-Before-Explain (predict-first protocol): by articulating prior ideas before instruction, learners create a cognitive 'hook' for new information. The side-by-side phenomenon images maintain the storyline thread and prime learners to think causally (not just descriptively) about transpiration.	Observable indicator: Every learner has at least one written prediction in their notebook within 2 minutes. Teacher scans notebooks during the cold-call phase. If fewer than half the class can generate any prediction, pause and ask a scaffold question: 'Think about what happens to YOUR body when you sweat on a hot day in Nairobi — does sweating help you or hurt you?' This connects to prior knowledge from respiration and homeostasis contexts.
Observe Phase  VIDEO: Hydrogen bonding in water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners work in groups of 4–5. Each group receives a Resource Pack containing: (i) a printed summary data table from Lessons 4–5 showing how stomata count, wind, temperature, and humidity affected transpiration rates in the potometer experiments; (ii) a short illustrated text (one A4 page,	1. Before distributing packs, say: 'In the last six lessons you gathered evidence. Now we are going to use that evidence plus new information to BUILD an explanation — like a scientist does. Scientists do not just collect data; they explain WHY the data means what it means.' 2. Distribute	Evidence-Based Discussion with Annotated Text: learners physically mark text (underline/circle) to distinguish evidence from explanation and to flag confusion. This is a disciplinary literacy strategy (SEP: Obtaining, Evaluating, and Communicating Information) that	Observable indicator: each group's annotated resource pack shows at least 3 underlines (useful functions) and at least 1 circle (surprise/confusion). Teacher uses circled items as discussion seeds in Phase 3. If a group has no circles, prompt: 'Does everything here make complete sense to

 HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES for similar readings	bilingual Swahili/English) on the cohesion-tension theory, the cooling function of transpiration, and mineral salt distribution; (iii) a printed excerpt or teacher-summarised paragraph on the role of forests in water cycling, specifically referencing the Mau Forest catchment in the Rift Valley and Lake Victoria's water balance. Groups read, discuss, and annotate: they underline key evidence that answers 'Why is transpiration useful?' and circle anything that surprises or confuses them. Groups also examine the fertiliser packet labels (DAP, CAN, Urea) brought as realia (sealed; not opened) and identify which nutrients each provides to the maize plant.	Resource Packs. Give groups exactly 10 minutes to read and annotate. Set a visible timer. Circulate — spend no more than 2 minutes per group. Prompt with: 'What does the cohesion-tension theory tell us about WHY the red rings appeared only in the xylem?' (WAIT 12 seconds before offering a hint.) 3. At the 5-minute mark, call out: 'Make sure your group has identified at least THREE pieces of evidence from our earlier experiments that connect to transpiration being useful.' 4. At 10 minutes, call time. Ask each group to appoint a 'reporter' for the next phase. 5. Hold up the sealed DAP, CAN, and Urea packets. Ask: 'Before I tell you what these are — what do you already know about how farmers in the Rift Valley use these white powders on their maize?' Cold-call 3 learners. Record responses. Then confirm: 'These are fertilisers. DAP gives nitrogen and phosphorus; CAN gives calcium and nitrogen; Urea gives nitrogen. Today we will connect these to what the xylem and phloem do inside the maize plant.'	makes the evidence-to-explanation move visible. Connecting fertiliser realia to xylem/phloem transport maintains the phenomenon storyline and grounds the chemistry in the biology.	you? What would you want to ask a Kenyan botanist if she walked in right now?'
Explain Phase  VIDEO: Hydrogen bonding in water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES	Groups share their findings in a structured whole-class discussion. The teacher co-constructs a 'Transpiration Functions Wall' on the board — three columns: (1) Cooling of Leaves, (2) Transpiration Pull / Cohesion-Tension, (3) Mineral Salt Distribution. Learners contribute evidence and explanations; the teacher scribes, questions, and refines. Learners then individually write a 3-sentence 'scientist's explanation' in their notebooks linking the phenomenon (red dye	1. Draw the three-column 'Transpiration Functions Wall' on the board. Say: 'We are going to BUILD this wall together using YOUR evidence and the new reading. Each brick in this wall must be supported by evidence — not just opinion.' 2. Column 1 — Cooling: Ask reporter from Group 1: 'What did your reading say about why a maize leaf does not overheat on a hot afternoon in Kisumu?' After response, ask the class: 'Do you AGREE or can you add evidence? Think back to our	Co-Constructed Concept Wall (collaborative explanation-building): rather than the teacher delivering a lecture, the board serves as a shared artefact that learners contribute to brick-by-brick. This makes explanation-building a social, visible, iterative process (SEP: Constructing Explanations). Anchoring each column back to the phenomenon images (red dye; water droplets in bag) sustains the storyline and prevents abstraction from becoming disconnected from the	Observable indicators: (i) the Transpiration Functions Wall has at least 2 pieces of learner-generated evidence per column by end of discussion; (ii) individual 3-sentence explanations include at least one reference to the phenomenon (red dye OR water droplets) AND at least one named function of transpiration. Teacher collects 6 notebooks (2 high, 2 mid, 2 lower) to scan during the group work in Phase 4. If explanations are purely definitional (no evidence linkage), teacher

for similar readings	in xylem; water droplets in the bag) to at least TWO functions of transpiration. A second mini-discussion explores the Mau Forest connection: how transpiration from millions of trees feeds water back into the atmosphere and sustains rainfall patterns across western Kenya and the Lake Victoria basin.	potometer results — what happened to transpiration rate when temperature was high?' (WAIT 15 seconds.) Cold-call 2 more learners. Scribe key phrase: 'Evaporation of water from stomata absorbs latent heat → cools the leaf.' 3. Column 2 — Transpiration Pull: Ask reporter from Group 2: 'How does the cohesion-tension theory explain the RED RINGS we saw in the maize stem?' (WAIT 12 seconds.) If response is incomplete, scaffold: 'Water molecules hold onto each other — what is that called? And they hold onto the xylem wall — what is that called?' Build the explanation: 'Transpiration at the leaf surface → creates tension (negative pressure) → pulls the water column upward through xylem → this is the transpiration pull.' Point to the red-dye image: 'THIS is why the dye moved up. Without transpiration — no pull. Without the pull — the water stays in the beaker.' 4. Column 3 — Mineral Salt Distribution: Ask: 'DAP contains phosphorus and nitrogen. Once a farmer in the Rift Valley applies DAP around a maize plant, HOW does that phosphorus reach the topmost leaf?' (WAIT 15 seconds.) Cold-call 3 learners. Guide toward: 'Minerals dissolved in soil water → absorbed through root hair cells → carried upward in the xylem by the transpiration stream → distributed to all parts of the plant.' Say: 'So transpiration is not just water loss — it is the DELIVERY MECHANISM for mineral nutrients.' 5. Individual writing (3 minutes silent): 'In your own words, write 3 sentences explaining — using evidence from our experiments — why	original observations.	pauses Phase 4 and models one sentence: 'The red dye moved up through the xylem because transpiration at the leaf pulled the water column upward — this is the transpiration pull (cohesion-tension theory).'
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		transpiration is important for the maize plant.' 6. Mau Forest mini-discussion (5 minutes): Display a simple sketch map of Kenya showing the Mau Forest, Rift Valley rivers, and Lake Victoria. Say: 'The Mau Forest is Kenya's largest water tower. Millions of trees transpire every day. Where does all that water vapour go?' Cold-call 4 learners. Build toward: 'Water vapour → forms clouds → returns as rainfall → feeds rivers → feeds Lake Victoria → supports farming communities in Kisumu, Kericho, and the broader western Kenya region.' Ask: 'So when forests are cut down — like parts of the Mau — what happens to this cycle?' (WAIT 12 seconds.) Accept 3 responses. Say: 'This is why transpiration matters beyond the single maize plant — it is part of Kenya's water security.'		
Driving Question Board (DQB) Creation  VIDEO: Hydrogen bonding in water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front of the room. Learners individually write one sticky note (or a strip of paper) answering a specific sub-question of the DQB that this lesson has addressed. They also add at least one NEW question that Lesson 7 has raised for them — especially about what happens when transpiration is disrupted (e.g., drought, deforestation, over-fertilisation). Groups then do a 'gallery walk' past the updated DQB for 3 minutes, reading peers' additions and placing a tick on any idea they agree with or a question mark on anything they want to discuss further.	1. Uncover the DQB. Point to the driving question written at the top: 'How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?' Say: 'At the start of this unit, this question was a mystery. Look at how many of these smaller questions on our board we can now answer.' 2. Read aloud 3 previously posted learner questions from earlier lessons (choose ones directly addressed today — e.g., 'Why does water keep moving up?'; 'What is the point of losing water from leaves?'; 'How do minerals get to the top of the plant?'). For each, ask: 'Can we answer this now? Who wants to give a one-sentence answer?' Cold-call 1 learner per question.	Driving Question Board Revision (tracking cumulative understanding): the DQB functions as a living, public record of the class's growing understanding. Marking questions as 'EXPLAINED ✓' gives learners a concrete, visible sense of intellectual progress and motivates engagement with remaining mysteries. The gallery walk activates peer-to-peer evaluation (SEP: Obtaining, Evaluating, and Communicating Information).	Observable indicators: (i) every learner has posted at least one sticky note with a full explanatory sentence AND one new question; (ii) at least 3 previously posted DQB questions are formally marked 'EXPLAINED ✓' with a class-agreed answer. Teacher photographs the updated DQB for the ARES platform record and to inform Lesson 8 planning. If learners cannot articulate any answered question, this signals that Phase 3 explanation-building was insufficient — teacher notes this for differentiated support in Lesson 8.

		Confirm or refine the answer and mark those questions as 'EXPLAINED ✓' on the board. 3. Distribute sticky notes. Say: 'Write ONE thing you can now explain that you could not explain in Lesson 1. Write it as a full sentence. Then write ONE new question this lesson has opened up for you.' Allow 3 minutes silent writing. 4. Learners post their sticky notes on the DQB. Run the gallery walk (3 minutes). 5. After the gallery walk, ask: 'What is still unexplained on our board about the maize plant and water?' Accept 3 responses. Highlight any that will be addressed in Lesson 8 (final lesson — synthesis and summative assessment preparation). Say: 'These open questions are NOT failures — they are the frontier of your learning. Real scientists in Kenya working on drought-resistant maize varieties at KARI [Kenya Agricultural and Livestock Research Organisation] ask questions just like these every day.'		
Model Building Phase  VIDEO: Hydrogen bonding in water Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES	Part A — Model Revision (10 min): Each learner opens their cumulative Transport in Plants model (begun in Lesson 1 and revised each lesson). They add a new annotated layer specifically showing: (i) arrows for the transpiration stream from roots → xylem → leaves → stomata → atmosphere; (ii) a label showing where mineral salts (e.g., nitrogen from CAN, phosphorus from DAP) travel within the xylem stream; (iii) a small cloud symbol above the leaf with a dotted arrow back to the Mau Forest sketch, representing the water cycle link. Learners add	1. Model revision: Say: 'Your model has been growing every lesson. Today you are adding the FINAL missing piece — what happens to the water AFTER it reaches the topmost leaf. Add the transpiration stream, the mineral delivery route, and the link to the water cycle. Use the three colours in your key.' Circulate for 10 minutes. Prompt with: 'Where exactly do the mineral salts enter the xylem? Show me on your model.' (WAIT 10 seconds.) Check that models show minerals entering at the root and travelling upward — not downward. 2.	Cumulative Model Revision + Community Communication (SEP: Constructing Explanations + Communicating Information): the model revision makes the full transport pathway — from soil water uptake through to atmospheric water vapour — visible in a single annotated diagram, integrating all lesson evidence. The community awareness task is an authentic communication challenge that requires learners to translate disciplinary knowledge into accessible language, deepening their own understanding through	Observable indicators: (i) learner models show a continuous, annotated transpiration stream from root to atmosphere, with mineral salt pathway labelled and water cycle link included; (ii) community messages are comprehensible to a non-specialist, include at least one specific actionable tip, and reference at least one fertiliser by name (DAP, CAN, or Urea); (iii) class thumb-voting reveals whether peers judge messages as actionable — teacher uses this as a proxy for clarity of explanation. Teacher collects 4 models (one

for similar readings	a colour-coded key distinguishing water movement (blue), mineral movement (green), and water vapour / transpiration loss (red dashed). Part B — Community Awareness Message (15 min): In their groups, learners draft a short awareness message (maximum 80 words, bilingual Swahili/English) addressed to a smallholder maize farmer in the Kisumu region. The message must: (i) explain in simple terms why watering the maize regularly matters (transpiration pull needs a continuous water column); (ii) advise on manuring to improve soil water retention; (iii) give ONE specific, practical tip on responsible fertiliser use — choosing DAP, CAN, or Urea appropriately and avoiding over-application to prevent soil and water pollution. Groups present their messages to the class in a 60-second 'radio broadcast' style — one learner reads it aloud as if speaking on a community FM radio station.	Community message task: Say: 'Scientists do not just understand the world — they communicate that understanding to help their communities. A farmer in Kisumu growing maize to feed her family needs to know what YOU now know about transpiration, water, and fertilisers. You are going to write her a message — short enough to broadcast on a community FM station like Radio Lake Victoria.' Circulate during drafting. Prompt groups: 'Would a farmer who has not studied biology understand the word xylem? How would you say it differently?' Encourage use of Kiswahili terms where helpful (e.g., 'mfumo wa usafirishaji wa maji' — water transport system). 3. Radio broadcast presentations: Each group stands and delivers their 60-second broadcast. After each, ask the class: 'Did this message give the farmer ONE specific action they can take? Thumbs up / thumbs sideways / thumbs down.' Briefly affirm specific actions mentioned and gently redirect any message that is too abstract or too technical. 4. Close by returning to the driving question one final time: 'So — what happens to the water in the end?' Cold-call 3 learners for a complete one-sentence answer. Confirm: 'The water travels up through the xylem, pulled by the transpiration stream driven by evaporation at the leaf surface through stomata — and it returns to the atmosphere as water vapour, completing the water cycle that sustains rainfall across western Kenya.' Say: 'In Lesson 8, we will pull everything together and prepare to show what we	the act of explanation (the 'protégé effect'). The radio broadcast format is culturally familiar to Kenyan rural contexts and builds public-speaking confidence.	per ability band) for detailed written feedback before Lesson 8.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your ARES journal:

1. **EXPLANATION QUALITY:** Did learners' 3-sentence individual explanations (Phase 3) include evidence from the phenomenon — or were they purely definitional? If most explanations were definitional, plan a brief 5-minute model explanation at the start of Lesson 8 using the red-dye image explicitly.
2. **DQB MOMENTUM:** Were learners able to identify at least 3 questions from the DQB that this lesson answered? If not, consider whether the Phase 2 resource texts were pitched at the right reading level for your class in Kisumu. Simplify or pre-read aloud in Lesson 8 if needed.
3. **COMMUNITY MESSAGE QUALITY:** Did the radio broadcast messages strike a balance between scientific accuracy and accessibility? Note which groups produced the strongest messages and consider sharing these on the ARES platform as exemplars for other schools.
4. **EQUITY:** Did the same learners dominate whole-class discussion across all phases? Review your cold-call log and ensure you are distributing participation equitably — particularly across gender and across learners with different levels of prior attainment.
5. **DRIVING QUESTION PROGRESS:** The driving question asks what happens to the water 'in the end.' By the end of Lesson 7, can most learners articulate a complete answer — from soil, through root hair cells, up the xylem via transpiration pull, out through stomata, and back to the atmosphere? If not, Lesson 8 must include a synthesis activity (e.g., annotated poster or flow diagram) to consolidate this full pathway before summative assessment.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In earlier lessons, learners placed a maize seedling's roots in red-dyed water and observed vivid red rings appearing only in the xylem tissue of a cross-sectioned stem. They also observed water droplets forming on the inside of a sealed polythene bag enclosing a leafy plant cutting placed in afternoon sun — even though no liquid water was added to the bag.
What did I learn?	Transpiration serves three critical functions for the maize plant: (1) it cools the leaf surface through evaporation, preventing heat damage on hot Kenyan afternoons; (2) it generates the transpiration pull (cohesion-tension mechanism) that is the primary force driving water upward through the xylem against gravity — explaining why the red dye moved upward through those specific channels; and (3) it drives the continuous transpiration stream that carries dissolved mineral salts (such as nitrogen from CAN/Urea and phosphorus from DAP) from the soil, through the roots, and all the way to the topmost leaves. At the landscape scale, transpiration from forests like the Mau Forest feeds water vapour back into the atmosphere, sustaining rainfall patterns that support farming communities across western Kenya and the Lake Victoria basin.
How does this explain the phenomenon?	The water that entered the maize plant through its root hair cells — visible as the red dye moving through the xylem rings — ultimately escapes as water vapour through the stomata of the topmost leaves (the water droplets seen on the inside of the polythene bag). This loss is not waste: it is the engine of the entire transport system. Without transpiration pulling water upward through the cohesion-tension mechanism, mineral nutrients dissolved in the soil water could not be delivered to the growing tissues. Responsible farming practices — regular watering to maintain the water column, manuring to improve soil structure and water retention, and careful use of fertilisers (DAP for phosphorus and nitrogen, CAN and Urea for nitrogen) — directly support this transport system and sustain maize crop yields on Kenyan farms.

LESSON 8 (40 minutes): MODEL BUILDING AND DQB CLOSURE: The Full Journey of Water in a Maize Plant

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all learning from Lessons 1 through 7 by building a complete annotated model of water and mineral transport in a maize plant, closing the Driving Question Board, and demonstrating understanding through a performance task set in a Kenyan farming context
Knowledge	Students synthesise knowledge of osmosis, root pressure, cohesion-tension theory, transpiration pull, translocation, and the three functions of transpiration into one unified explanatory framework tracing a water molecule from soil to atmosphere
Skills	Students revise and annotate scientific diagrams, provide and receive structured peer feedback, evaluate which DQB questions have been answered and which remain open, and construct a written explanation for a novel scenario involving a wilting tomato plant in Mombasa coastal heat
Attitudes	Students demonstrate intellectual honesty by comparing their final models with their Lesson 1 predictions and acknowledging how their thinking has changed; they show responsibility by connecting plant transport knowledge to sustainable farming and water conservation in Kenya
Key Inquiry Question	Why is transport in plants important? — addressed fully by demonstrating that without an integrated system of absorption, xylem transport, translocation, and transpiration, a maize plant on a Kenyan farm could not absorb mineral nutrients, cool its leaves, grow, or contribute to the local water cycle
Purpose in Storyline	This is the capstone lesson. Students return to the original dyed-water maize seedling and polythene-bag phenomena from Lesson 1 with full explanatory power, revise their initial sketches into complete mechanistic models, close the Driving Question Board, and apply their understanding to a new Kenyan context — a wilting tomato in Mombasa — proving genuine transfer of learning
Safety Notes	No hazardous materials in this lesson. Ensure scissors used to cut paper diagrams are handled safely. If printed diagrams are distributed, ensure ink is dry. Supervise group movement during peer feedback gallery walk to prevent crowding.

B. LESSON OVERVIEW

Lesson 8 is the final consolidation lesson for Sub-strand 1.4. Students begin by individually revisiting their Lesson 1 prediction sketches and annotating them with seven lessons of accumulated understanding. They then share and receive structured peer feedback in small groups. The Driving Question Board is formally reviewed as a whole class: every posted question is evaluated — answered, partially answered, or still open. A short performance task (the wilting tomato scenario in Mombasa) serves as the summative check for transfer. The lesson closes with students articulating what changed in their thinking from Lesson 1 to Lesson 8, anchoring the storyline in the original phenomenon of red dye in maize xylem and water droplets inside the polythene bag.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Students retrieve their original Lesson 1 sketches from their science notebooks or from the	Display both original phenomena on the board simultaneously: a photograph of the red-dyed maize	The I-used-to-think / Now-I-know routine activates metacognitive reflection and makes prior	Listen to the I-used-to-think / Now-I-know statements. Check whether students are identifying specific

Osmosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12 Search ARES for similar readings	class prediction wall. Working individually for 4 minutes, each student silently compares their original sketch with what they now know. They write a brief personal note beside the sketch: what did I think then, and what do I know now? They identify at least two things their original sketch got wrong or left out. This primes retrieval of all prior learning before model revision begins.	stem cross-section showing vivid red rings in the xylem, and a photograph of water droplets on the inside of the polythene bag from the succulent cutting. Say: 'Before you open your notebooks, look at these two images that started our entire journey eight lessons ago. Without talking to anyone, take out your Lesson 1 sketch and read what you wrote. Do not change anything yet — just read it as if you are meeting a younger version of yourself.' Allow 90 seconds of silent reading. Then say: 'Now write one sentence next to your old sketch that begins with the words: I used to think... and one sentence that begins: Now I know...' WAIT TIME: Allow 2 full minutes of silent writing before any student shares. Do not prompt or hint — this is individual retrieval time. After writing, cold-call two or three students to share their sentences aloud. Affirm all responses: 'That shift in thinking is exactly what scientific learning looks like.'	knowledge explicit before revision. Displaying the original phenomena reconnects students to the anchoring experience that opened the unit, creating narrative closure. The silence enforces individual thinking before social construction begins.	mechanisms (osmosis, cohesion-tension, stomata) or remaining vague (water goes up). Students who remain vague will need targeted support during the model-building phase. Note names for follow-up questioning.
Observe Phase  VIDEO: Osmosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12 Search ARES for similar readings	Students observe a reference set of four visual anchors displayed at the front or on printed handouts: (1) a labelled cross-section of a maize stem showing xylem and phloem positions — linking to why only the xylem rings turned red; (2) a diagram of a root hair cell showing osmosis and active transport of mineral ions; (3) a graph pooled from Lesson 6 showing how wind, humidity, and light affect transpiration rate — the class-generated data; (4) a sketch of the Mau Forest water cycle showing transpiration feeding into regional rainfall. Students spend 3 minutes in silence annotating	Distribute or project the four reference images. Say: 'These four images summarise the evidence we have collected and analysed across seven lessons. For the next three minutes, annotate each one with a brief label that connects it to our driving question. You are not building your model yet — you are just reminding yourself what evidence you have.' WAIT TIME: Allow 3 full minutes of silent annotation. Circulate and observe — do not answer questions yet. After 3 minutes say: 'Now turn to the person next to you. In 60 seconds, tell them which of these four pieces of evidence you find	The four-image evidence review functions as a visual summary of the entire inquiry arc. Pairing it with a directed annotation task ensures students engage actively rather than passively viewing. The pair-share on most powerful evidence develops scientific argumentation — students must justify their choice, not just identify it.	During the silent annotation walk, note which images students skip or annotate only superficially. Students who skip the root hair osmosis image may have gaps in understanding absorption mechanisms. Students who annotate the Mau Forest image only vaguely may not have connected transpiration to the water cycle. These gaps inform model-building support.

	these four images with sticky labels or pencil notes linking each image back to the driving question: 'How does a maize plant move water and nutrients from soil to its topmost leaves — and what happens to that water in the end?' They are observing their own cumulative evidence before constructing the final model.	most powerful in answering the driving question, and why.' After the pair-share, ask two pairs to report. Restate key mechanisms using precise language: 'So the cohesion-tension mechanism — where water molecules pull each other upward through the xylem because they are attracted to each other and to the xylem walls — is the engine behind what we saw in that red-dyed maize stem.'		
Explain Phase  VIDEO: Osmosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12  Search ARES for similar readings	Students individually revise their Lesson 1 sketch into a complete annotated diagram over 8 minutes. The revised diagram must trace one water molecule on its complete journey: soil solution entering root hair cell by osmosis, crossing the cortex, entering xylem in the root, travelling up the xylem through the stem to the leaf (driven by cohesion-tension and transpiration pull), exiting through stomata as water vapour during transpiration. Each arrow in the diagram must carry a mechanism label: osmosis, active transport for mineral ions, cohesion-tension, transpiration pull, evaporation. At least one annotation must explain why the xylem — not the phloem — turned red in the maize stem. At least one annotation must explain where the water droplets on the polythene bag came from. Students then share diagrams in groups of three and give each other one piece of warm feedback (something explained well) and one piece of cool feedback (something to add or clarify), using the sentence frames: 'Your diagram clearly shows... because...' and 'One thing to add or strengthen is... because...' Groups have 4 minutes for peer	Say: 'You now have 8 minutes to transform your Lesson 1 sketch into the most complete, most accurate diagram you have ever drawn of water movement in a plant. Every arrow must have a mechanism. Every step must link back to at least one observation or experiment we did together.' WAIT TIME: Allow 8 full minutes of silent individual work. Resist the urge to lecture during this time — circulate, crouch to student level, read diagrams, and place a small tick near strong annotations to affirm without interrupting. After 8 minutes, instruct groups of three to form. Say: 'You will give your neighbour exactly one warm comment and one cool comment. Use the sentence frames on the board. You have 4 minutes — 90 seconds per person, then rotate.' After peer feedback, ask one group to share the most interesting cool feedback they received: 'What did your peer suggest you add, and do you agree?' If a student says they were told to add the role of guard cells in controlling stomatal opening, affirm: 'Excellent — guard cells are the gatekeepers of transpiration. If a farmer in Kisumu sees wilting in the afternoon, it is those guard cells	Requiring mechanism labels on every arrow prevents students from drawing a pretty diagram without understanding. The warm-cool peer feedback protocol develops scientific communication and forces students to evaluate each others reasoning, not just appearance. Connecting guard cells to a Kisumu farmer grounds abstract plant physiology in lived Kenyan experience.	Collect or photograph three to five diagrams from students who showed gaps in the Observe Phase. Check specifically: Does the diagram show osmosis at the root hair? Does it distinguish xylem (water) from phloem (sugars)? Does it explain transpiration pull? Is the fate of water (exit through stomata as vapour) shown? Use these as exit data to identify students needing support before the next topic.

	feedback.	closing stomata to conserve water.' This connects mechanism to Kenyan farming context.		
Driving Question Board (DQB) Creation VIDEO: Osmosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Osmosis (PLIX) Source: CK-12 Search ARES for similar readings	The full class gathers facing the Driving Question Board that has been building since Lesson 1. The teacher reads out each posted question one by one. Students vote using a three-colour card system: green card means fully answered, yellow card means partially answered, red card means still open or unanswered. After each vote, one student who voted green must state the answer in one sentence. If a question gets mostly red cards, the class briefly discusses why it remains open and whether it points to a new inquiry. The DQB review takes approximately 5 minutes. At the end, the teacher formally closes the board by placing a large green star on the driving question: 'How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?' and asks the class to chorus the one-sentence answer together.	Stand at the DQB with a green marker. Read each question clearly and pause before the vote. After green votes, use the cold-call: 'Achieng, give us the one-sentence answer for that question.' After the vote on any red-card question, say: 'This question is still open — that is not a failure. Real scientists work with open questions every day. Write it in your notebook under: Questions I still have about plants.' When the board reaches the driving question, say: 'This was our compass for eight lessons. Look at where we started — sketches and guesses. Look at where we are now — mechanisms, experiments, graphs, and community applications. Let us say the answer together.' Lead the class in a choral one-sentence answer. Suggested answer to guide: 'A maize plant absorbs water by osmosis through root hairs, pulls it up the xylem by cohesion-tension and transpiration pull, delivers mineral salts by active transport, and releases water vapour through stomata — cooling the leaf, sustaining the transpiration stream, and feeding water back into the atmosphere over farms and forests like the Mau.' WAIT TIME: After the choral answer, allow 20 seconds of silence before moving to the performance task. Let the closure land.	The three-colour card voting makes every student publicly commit to an evaluation of each question — no passive observers. Requiring green-voters to state the answer shifts accountability from the teacher to the learner. The choral final answer creates a shared moment of closure that is emotionally and cognitively satisfying, reinforcing the narrative arc of the storyline.	The pattern of green, yellow, and red cards across questions reveals class-wide gaps. If the question about phloem translocation gets mostly yellow or red, this signals that Lesson 4 content needs revisiting in the next topic review. Record the distribution of responses for the three most conceptually demanding questions in the teacher reflection log.
Model Building Phase VIDEO:	Students complete the performance task individually in the final 6 minutes. The task prompt is displayed and read	Display the performance task prompt in large text. Read it aloud once. Say: 'This is a new plant, a new city, a new farmer — but the	The novel-scenario performance task assesses transfer — the highest cognitive demand — by requiring students to apply	Collect the performance task responses as the primary summative assessment for Sub-strand 1.4. Score against four

Osmosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Osmosis (PLIX) Source: CK-12 Search ARES for similar readings	aloud: 'A tomato farmer in Mombasa notices that her plants are wilting badly at 2 pm on a hot, humid afternoon. She waters them at 6 pm, and by 8 pm the plants have recovered and look healthy again. Using your knowledge of osmosis, transpiration, stomata, and the transpiration pull, write a scientific explanation of (a) why the plant wilted at 2 pm, (b) what happened inside the plant cells when she watered at 6 pm, and (c) why the coastal humidity during the day did not prevent wilting. Your explanation must name at least four mechanisms from our unit.' Students write their responses in their science notebooks. After writing, two or three volunteers read their explanations aloud and the class identifies which mechanisms were named correctly.	science is the same science we have been building since Lesson 1. You have everything you need. Write your explanation now.' WAIT TIME: Allow 5 full minutes of uninterrupted individual writing. Do not circulate during this time — let students work independently without the scaffold of teacher proximity. After 5 minutes, ask for two volunteers. After the first volunteer reads, ask the class: 'Which four mechanisms did you hear?' List them on the board as students call them out. After the second volunteer, say: 'Did the second explanation add anything the first missed?' Close the lesson by returning to the original images: 'The red rings in that maize stem. The water droplets on that polythene bag. You now have the science to explain both — and to explain a wilting tomato in Mombasa, and to advise a farmer in Kisumu on when to water and which fertiliser to use. That is the power of understanding transport in plants.' Thank students by name for contributions across the eight lessons.	mechanisms to an unfamiliar organism (tomato), location (Mombasa coast), and time condition (high afternoon heat plus coastal humidity). The requirement to name four mechanisms prevents surface-level narrative and demands mechanistic scientific explanation. Closing with the original phenomena images creates full narrative closure for the storyline.	criteria: (1) correct use of osmosis to explain wilting and recovery; (2) correct explanation of stomatal closure under water stress at 2 pm; (3) correct use of cohesion-tension or transpiration pull; (4) correct explanation of why high humidity slows but does not stop transpiration when leaf water potential is very low due to heat stress. Students scoring below 2 out of 4 criteria should receive targeted consolidation support before Sub-strand 1.5.
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D. TEACHER REFLECTION

After the lesson, consider: Did the I-used-to-think / Now-I-know statements reveal genuine conceptual shift, or did students simply repeat correct vocabulary without understanding? Were students able to distinguish xylem from phloem functions in their revised diagrams without being prompted? Did the Mombasa tomato performance task reveal any persistent misconceptions — for example, did students confuse osmosis with active transport, or did any student claim that high coastal humidity would completely stop transpiration? Which DQB questions received the most red cards, and what does that reveal about pacing or depth of coverage in Lessons 3 and 4? How effectively did the peer feedback protocol push students to deepen their diagrams, or did warm feedback dominate and cool feedback feel uncomfortable? Consider whether Kenyan contexts — Kisumu farms, the Mau Forest, Mombasa coastal conditions, DAP and CAN fertiliser use — felt authentic to students or were experienced as superficial additions. Plan to carry any still-open DQB questions forward as entry points into Sub-strand 1.5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed red dye moving upward through the xylem rings of a cut maize stem cross-section, water droplets forming on the inside of a polythene bag enclosing a leafy cutting, transpiration rates changing with wind speed, humidity, and light intensity in our group experiments, and wilting and recovery in plants under water stress — all real phenomena visible in Kenyan school gardens and farms.
What did I learn?	We learned that water enters root hair cells by osmosis down a water potential gradient, that mineral salts enter by active transport requiring energy, that water travels up the xylem — not the phloem — driven by cohesion-tension and transpiration pull created at the leaf surface, that sugars travel downward and upward through the phloem by translocation from source to sink, that transpiration cools leaves, sustains the transpiration stream carrying minerals to the topmost leaves, and contributes water vapour to regional water cycles such as those sustained by the Mau Forest.
How does this explain the phenomenon?	We explained why only the xylem rings turned red in the maize stem cross-section, why water droplets formed on the polythene bag from evaporation through stomata, why a wilting plant in Mombasa recovers after evening watering through osmosis restoring cell turgor, why DAP fertiliser applied to a Kisumu maize farm reaches the topmost leaves via the transpiration stream, and why cutting a ring of bark around a stem (girdling) kills the plant above the cut by blocking phloem translocation while leaving xylem intact.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.